



# Aero-thermal optimization of a gas turbine blade endwall with non-axisymmetric contouring and purge flow

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## ABSTRACT

With the ever-increasing turbine inlet temperature and blade loadings, the thermal management for the endwall of gas turbines is gaining wide attention. Specifically, the increasing blade loadings intensify the cross flows and the blade-endwall interactions, making the design of turbine endwalls operating in such harsh thermal environments a really challenging problem. To resolve such a problem, we propose a novel non-axisymmetric endwall, which is shaped both near the blade leading edge and between the two adjacent blades. The effectiveness of the proposed non-axisymmetric endwall is well demonstrated on a typical large-turning-angle gas turbine with a purge slot. More specifically, in order to obtain a solution that can achieve a good balance between the gas turbine aerodynamic and thermal performance, we carry out three separate optimizations for the non-axisymmetric endwall. The aerodynamic loss, the film cooling effectiveness and the heat transfer coefficient are taken as the objective functions, respectively, for the optimization process. The optimal solution aiming to minimize the heat transfer coefficient achieves the most desirable performance. This solution substantially reduces both the aerodynamic loss and thermal loads at the turbine endwall, since it is effective to suppress both the horseshoe vortex and the cross flows. Furthermore, through a fundamental flow-thermal analysis of the three optimal solutions, an insight into the effects of non-axisymmetric endwall on gas turbine aero-thermal performance is obtained and guidelines for the aero-thermal design of non-axisymmetric endwalls are provided.

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## 1. Introduction

In recent years, the huge energy needs and the emphasis on environmental protection issues have led to greater demands on the efficiency and power of gas turbines that are broadly utilized for electricity generation and aerospace propulsion. In terms of the thermal cycle, higher gas turbine rotor inlet temperatures (RIT) can effectively enhance the thermal efficiency and output power of gas turbines. The current advanced gas turbines can operate at a RIT of 1700°C, which far exceeds the melting point of blade materials [1]. Besides, modern gas turbines have updated the diffusion burners to premixed burners to minimize NO<sub>x</sub> emissions. The radial temperature profile of the hot gas at the turbine entrance is thereby flatter, causing a higher temperature near blade endwalls accordingly. With increasingly high heat loads being imposed on gas turbine endwalls, the thermal management for turbine endwalls is receiving increasing attention. Unfortunately, the complex secondary flow structures near the endwall lead to a very harsh flow and thermal environment there, posing a serious challenge to

the endwall thermal management [2]. Therefore, there is an urgent necessity to develop state-of-the-art flow control technologies and efficient cooling schemes for the designs of modern gas turbine endwalls.

The endwall secondary flows, as typical three-dimensional flow features within a turbine passage, influence both blade aerodynamic performance and the thermal management for blade endwalls. Some noted scholars were among the pioneers to summarize the endwall secondary flow structure models [3–5]. They proposed classic models to describe the formation, growth and interaction of the secondary flows based on the insightful understanding of secondary flow mechanisms. Denton [6] pointed out that the secondary flows caused the mixing loss with the mainstream and the friction loss within the near-endwall boundary layer, thereby leading to significant aerodynamic losses of a turbine passage. Sharma et al. [7] quantitatively stated that endwall secondary flow losses may account for one-third of the overall aerodynamic losses of turbine stages. Blair [8] performed a pioneering study on endwall heat transfer and revealed its correlation with the near-endwall secondary vortices. Papa et al. [9] also discovered that the potential high heat transfer zones were strongly linked to the secondary flow structures. Simon and Piggush [10] emphasized that it was

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**Nomenclature**

|             |                                                  |
|-------------|--------------------------------------------------|
| $C$         | blade chord, mm                                  |
| $D$         | slot position, mm                                |
| $H$         | maximum amplitude, mm                            |
| $L$         | slot length, mm                                  |
| $S$         | blade span, mm                                   |
| $P$         | blade pitch, mm                                  |
| $W$         | slot width, mm                                   |
| $C_{ax}$    | axial chord, mm                                  |
| $C_p$       | static pressure coefficient                      |
| $L_0$       | upstream length, mm                              |
| $L_1$       | downstream length, mm                            |
| $q_w$       | heat flux, W/m <sup>2</sup>                      |
| $p_t$       | total pressure, Pa                               |
| $T_c$       | purge flow inlet temperature, K                  |
| $T_\infty$  | mainstream inlet temperature, K                  |
| $T_w$       | non-adiabatic wall temperature, K                |
| $T_{aw}$    | adiabatic wall temperature, K                    |
| $T_f$       | fluid temperature, K                             |
| $W_{ss}$    | suction side width, mm                           |
| $W_{ps}$    | pressure side width, mm                          |
| $Nu$        | Nusselt number                                   |
| $h$         | heat transfer coefficient, W/(m <sup>2</sup> ·K) |
| $Re_{exit}$ | exit Reynolds number                             |
| $Ma_{exit}$ | exit Mach number                                 |
| $MFR$       | mass flow ratio of coolant to mainstream         |

**Greek letter**

|              |                                       |
|--------------|---------------------------------------|
| $\alpha$     | slot orientation, deg                 |
| $\eta$       | adiabatic film cooling effectiveness  |
| $\lambda$    | thermal conductivity, W/(m·K)         |
| $\rho$       | density, kg/m <sup>3</sup>            |
| $\varphi$    | non-dimensional temperature           |
| $\phi$       | non-dimensional metal temperature     |
| $\eta_{ave}$ | averaged film cooling effectiveness   |
| $\xi_{p,t}$  | total pressure loss coefficient       |
| $\mu_L$      | laminar viscosity, kg/m·s             |
| $\mu_T$      | turbulent viscosity, kg/m·s           |
| $\delta$     | velocity boundary layer thickness, mm |
| $\theta$     | momentum boundary layer thickness, mm |

**Subscripts**

|          |                          |
|----------|--------------------------|
| $c$      | purge flow               |
| $t$      | stagnation condition     |
| $w$      | wall                     |
| $f$      | fluid                    |
| $\infty$ | main flow                |
| $aw$     | adiabatic wall condition |
| $ave$    | averaged values          |
| $ss$     | suction side             |
| $ps$     | pressure side            |
| $exit$   | blade exit               |

essential to sort out the association between the secondary flow patterns and the heat transfer and cooling behaviors of the turbine endwall.

Another typical feature of the endwall is the purge flow originating from the gaps at the combustor-turbine or stator-rotor interfaces. The purge flow can seal these gaps and cool the endwall surface as well [11]. The thermal impact of the purge flow on blade endwalls has attracted much interest over the past decade. Thole and Knost [12] pointed out that a purge flow could bring substantial film cooling benefit to blade endwalls, however, it was unfavorably influenced by the secondary flows. Papa et al. [2] and Card-

well et al. [13] experimentally reported that the purge cooling effectiveness and coolant spreading area were both enhanced at an elevated coolant-to-mainstream mass flow ratio (MFR). Du et al. [14] documented that the tangential ejection angle significantly influenced the cooling effectiveness of the endwall and vane suction surface. Liu et al. [15] found that increasing the density ratios (DR.) of the coolant relative to the mainstream could also enhance the cooling performance as heavier coolant could transport much nearer to the endwall surface. Several researchers also focused on the geometrical influence of the purge slot. Thrift et al. [16] experimentally revealed the influences of the purge slot's orientation and location on the purge cooling effectiveness of a vane endwall. Both of these two factors were demonstrated to significantly influence the film cooling efficiency. Tao et al. [17] reported that a purge slot with contracted cross-sectional geometry could accelerate the coolant to dramatically improve the cooling efficiency. In addition to the film cooling aspect, the influence of the purge flow on endwall heat transfer should not be underestimated. Based on experiments, Lynch and Thole [18] claimed that a purge flow considerably modified the endwall heat transfer behaviors due to its interactions with the complex near-endwall secondary flows. Kim et al. [19] demonstrated that the purge leakage intensified the heat transfer of the upstream endwall because of its turbulent mixing with the main flow.

With modern gas turbine designs evolving towards higher blade loadings, the secondary flows within the blade passage have become stronger. As established above, the presence of secondary flows makes the aerodynamic design and thermal management of the turbine blade a huge challenge. As a promising flow control technique, non-axisymmetric endwall contouring (NEC) has attracted extensive research interest in the last two decades. A number of pioneering researchers have devoted efforts to unravel the fundamentals of NEC and have focused primarily on its impact on aerodynamic performance [20–24]. The basic principle of NEC lies in altering the velocity and static pressure distribution near the endwall by forming a concave or convex curvature on the endwall surface. In general, the conventional NEC formed protrusions near the pressure side to accelerate the fluid there and formed valleys near the suction side to decelerate the fluid there. In such cases, the NEC helped to reduce the lateral pressure gradient within a blade passage, thus inhibiting the formation and development of secondary flows. The modeling methods for NEC can be generally classified into two categories according to their degrees of freedom. The first method used two pre-defined control curves in the axial (usually decay functions) and circumferential (usually shape functions) directions to fit the NEC surfaces with fewer design variables [20–22]. While the second method could create more complex NEC surfaces by means of cubic spline surfaces at the cost of more design variables [23,24].

In recent years, with the emphasis on thermal protection for modern gas turbines, the application of NEC in endwall thermal management has gained wide attention. Saha et al. [25] were pioneers to use the NEC to weaken the secondary flows and reduce the endwall heat transfer. They used two a product of two control curves to generate a profiled endwall surface, which resulted in an 8% reduction in the area-averaged heat transfer rates. Panchal et al. [26] stated that the endwall profiles obtained by optimizations for thermal and aerodynamics purposes both could lower the averaged heat transfer rate by not less than 10% at transonic exit conditions. Winkler et al. [27] utilized a novel approach called the ice-forming method to generate the NEC. They reported that the contoured endwall dramatically influenced the flow-thermal environments and vortex transports near the endwall, reshaping the heat transfer distributions on the endwall. Lynch et al. [28] experimentally documented that using the NEC could weaken the passage vortex, thereby reducing the heat transfer rates in high-heat-

**Table 1**  
Blade geometrical parameters.

| Parameters                          | Value  |
|-------------------------------------|--------|
| Axial chord, $C_{ax}$ / mm          | 129.64 |
| True chord, $C$ / mm                | 184.15 |
| Blade span/axial chord, $S/C_{ax}$  | 3.52   |
| Blade pitch/axial chord, $P/C_{ax}$ | 1.07   |
| Inlet angle/deg                     | 35     |
| Exit angle/deg                      | -72.5  |

transfer regions by around 20%. Mensch et al. [29] used the conjugate heat transfer simulations to evaluate the influence of NEC on the overall cooling effectiveness of the endwalls. They found that the profiled endwall had an adverse influence on the film cooling effectiveness and the impingement cooling effectiveness. Chen et al. [30] conducted a global sensitivity analysis to explore how the NEC influenced the aero-thermal performance of the endwalls and also determined the most significant parameters. Roy et al. [31] experimentally stated that for a transonic blade with upstream leakage flows, the adoption of NEC could effectively lower the heat loads to the endwalls. Tao et al. [32] investigated the influence of the NEC generated merely near the blade LE on aerodynamics and thermal management of a turbine endwall with a purge leakage. It was reported that the NEC significantly decreased the heat load of the juncture of the endwall and the blade LE.

Up to now, most studies on non-axisymmetric endwalls have not considered the upstream purge flow. As established above, the purge flow not only brings film cooling benefits but also interacts very strongly with the secondary flow structures [31,32]. Therefore, it is essential to investigate the NEC with consideration of the purge flow. Besides, to our best knowledge, the published literature usually focuses only on the control of the cross flows [25–31] or only on the control of the horseshoe vortex [32]. There is a lack of NEC designs intended to control both the horseshoe vortex and cross flows in the open literature. Therefore, a novel and feasible full-endwall contouring method aiming to weaken both the horseshoe vortex and cross flows is proposed in this paper. The effectiveness of the proposed NEC is demonstrated on a gas turbine endwall with a typical purge slot. Furthermore, the design of NEC often involves multiple performance objectives, however, most studies in the field of NEC have only focused on one or at most two objectives. To better find the desirable solutions with well-balanced aerodynamic and thermal performance, three optimizations with the objectives of the aerodynamic loss, the film cooling effectiveness and the heat transfer rate are performed for the NEC, respectively. At last, through an in-depth analysis of flow-thermal details, the influence of these optimal NEC designs on end-wall aero-thermal performance are clarified and the helpful design guidelines for the NEC are also suggested.

## 2. Numerical method and validation

### 2.1. Baseline computational model

Fig. 1 plots the geometry and computational domain of the baseline model with a flat endwall and a purge slot configuration. The baseline flat endwall is used as a benchmark for the optimization of non-axisymmetric endwalls. The upstream purge slot with a coolant supply chamber is constructed to generate the purge flow to cool the endwall. The baseline model's geometries are derived from the published literature of Papa et al. [2], which presents detailed experimental data. Fig. 1(b) and (c) define the geometrical parameters of the blade and the purge slot, respectively. Their detailed parameter values are presented in Tables 1 and 2, respectively.

**Table 2**  
Purge slot geometrical parameters.

| Parameters                  | Value |
|-----------------------------|-------|
| Orientation, $\alpha$ / deg | 45    |
| Width, $W$ / mm             | 4.00  |
| Length, $L$ / mm            | 25.50 |
| Position, $D$ / mm          | 12.96 |

**Table 3**  
Boundary conditions.

| Parameters        | Value            |
|-------------------|------------------|
| $\delta$ / mm     | 22               |
| $\theta$ / mm     | 2.2              |
| $Re_{exit}$       | 600,000          |
| $Ma_{in}$         | 0.045            |
| $Ma_{exit}$       | 0.14             |
| $MFR$             | 0.5%, 1.0%, 1.5% |
| $T_{c,total}$ / K | 298.15           |
| $T_{m,total}$ / K | 308.15           |

Table 3 summarizes the computational boundary conditions, which keep the same with the experimental setup of Papa et al. [2]. There is a general consensus that the boundary layer thickness at the mainstream entrance has a significant impact on endwall heat transfer predictions. Therefore, a 2-D boundary layer simulation is firstly performed to obtain the same velocity and momentum boundary layer thickness ( $\delta$  and  $\theta$ ) as in the experiment of Papa et al. [2]. Then the distributions of velocity, turbulent kinetic energy and turbulent specific dissipation along the span-wise direction are set as the mainstream inlet boundary conditions. At the chamber entrance, the coolant's mass flow rate is given based on the MFR value. To establish a single-passage computational domain, the translational periodic conditions are imposed at the two interfaces in the pitch-wise direction. To gain a complete and profound insight into the endwall aero-thermal performance, two simulations are performed for each design, with different thermal boundary conditions on the endwalls. In the case of purge cooling simulation, the endwalls are treated as adiabatic no-slip walls. In the case of heat transfer simulation, the endwalls are set as heat transfer walls with a constant heat flux of  $q_w=1000 \text{ W/m}^2$  according to Lynch and Thole [18].

### 2.2. Turbulence model validation

The multi-block structured meshes are generated to discretize the computation domains utilizing the commercial software NUMECA 9.1-2. Fig. 2 provides the computational mesh details for the baseline endwall and a typical contoured endwall. To obtain a high-quality mesh, there is an O-type topological structure for the grids around the blade [33]. A grid refinement is employed for all solid walls to better resolve the flow details within the boundary layer. Note that the meshes around the slot exit are also refined to capture the strong mixing of the coolant with the mainstream. During the turbulence model selection, the thickness of the near-wall grid is finely adjusted to suit the  $y^+$  values of various turbulence models. The commercial software ANSYS CFX 14.5 is employed to solve the steady Reynolds-Averaged Navier–Stokes (RANS) equations. The theory and modeling methods for the CFX solver can be found in Refs. [34,35]. The element-based finite volume method is adopted for spatial discretization. A point implicit linear equation solver coupled with an algebraic multigrid method is used to resolve the discretized governing equations. The grid sensitivity analysis is also carried out to determine a suitable mesh size. The mesh sizes of the baseline model (flat endwall) are varied to confirm the independence of the mesh setup. As shown in Fig. 3,

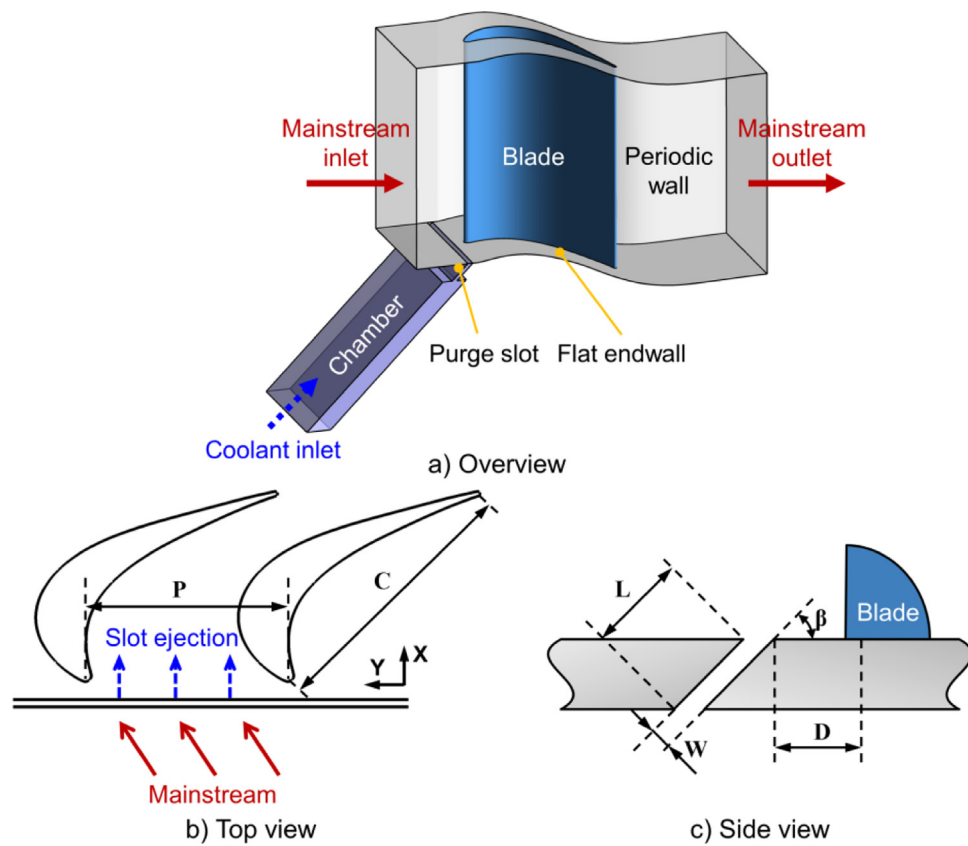


Fig. 1. Geometry and computational domain of the baseline model.

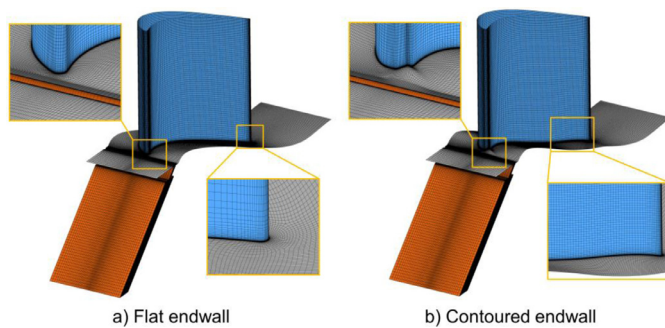


Fig. 2. Computational mesh for flat and contoured endwalls.

the area-averaged film cooling effectiveness (FCE) of the endwall is gradually increased as the grid number increases. When the grid number exceeds 4.3 million, the grid-independent results can be achieved. Therefore, a 4.9 million mesh is finally employed for the balance of the computational cost and accuracy.

Before the optimization, to verify the numerical method and assure the credibility of the predicted results, there is a need to select a suitable turbulence model. The numerical simulations are conducted with four widely-used turbulence models, including standard  $k-\varepsilon$ , standard  $k-\omega$ , RNG  $k-\varepsilon$ , and SST  $k-\omega$  model. Fig. 4 compares the endwall film cooling outcomes from the numerical predictions and the experiments of Papa et al. [2] at a blowing ratio of 0.5. Fig. 4(a) plots the film cooling effectiveness contours and Fig. 4(b) presents a quantitative comparison. As depicted in Fig. 4(a), the standard  $k-\varepsilon$  model is found to heavily overestimate the diffusion area of the purge coolant. On the contrary, there is an underestimation of film cooling coverage on the forepart of end-

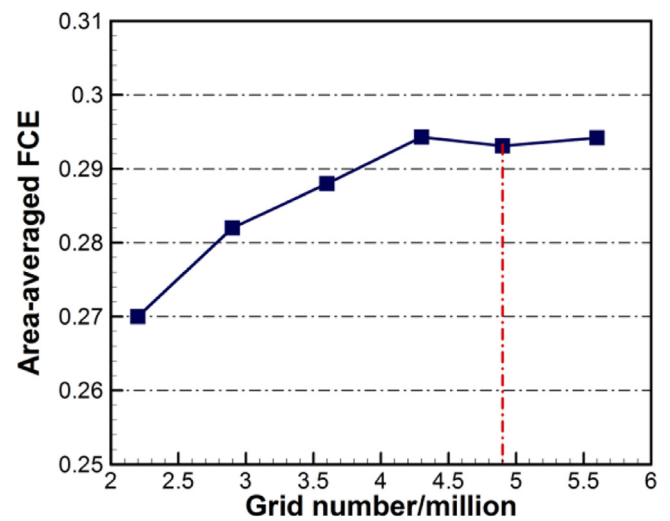


Fig. 3. Mesh independence test.

wall for the RNG  $k-\varepsilon$  and SST  $k-\omega$  models. Overall, the film cooling distribution predicted by the standard  $k-\omega$  model achieves the best agreement with the experiment results. Fig. 4(b) further compares the transversely averaged FCE. The standard  $k-\omega$  model is found to achieve the best quantitative prediction among the four turbulence models. This model can accurately predict the averaged film cooling effectiveness distributions for almost the entire endwall, with a maximum relative error below 4.8%. Taken together, the standard  $k-\omega$  model can accurately predict the endwall film cooling behaviors.



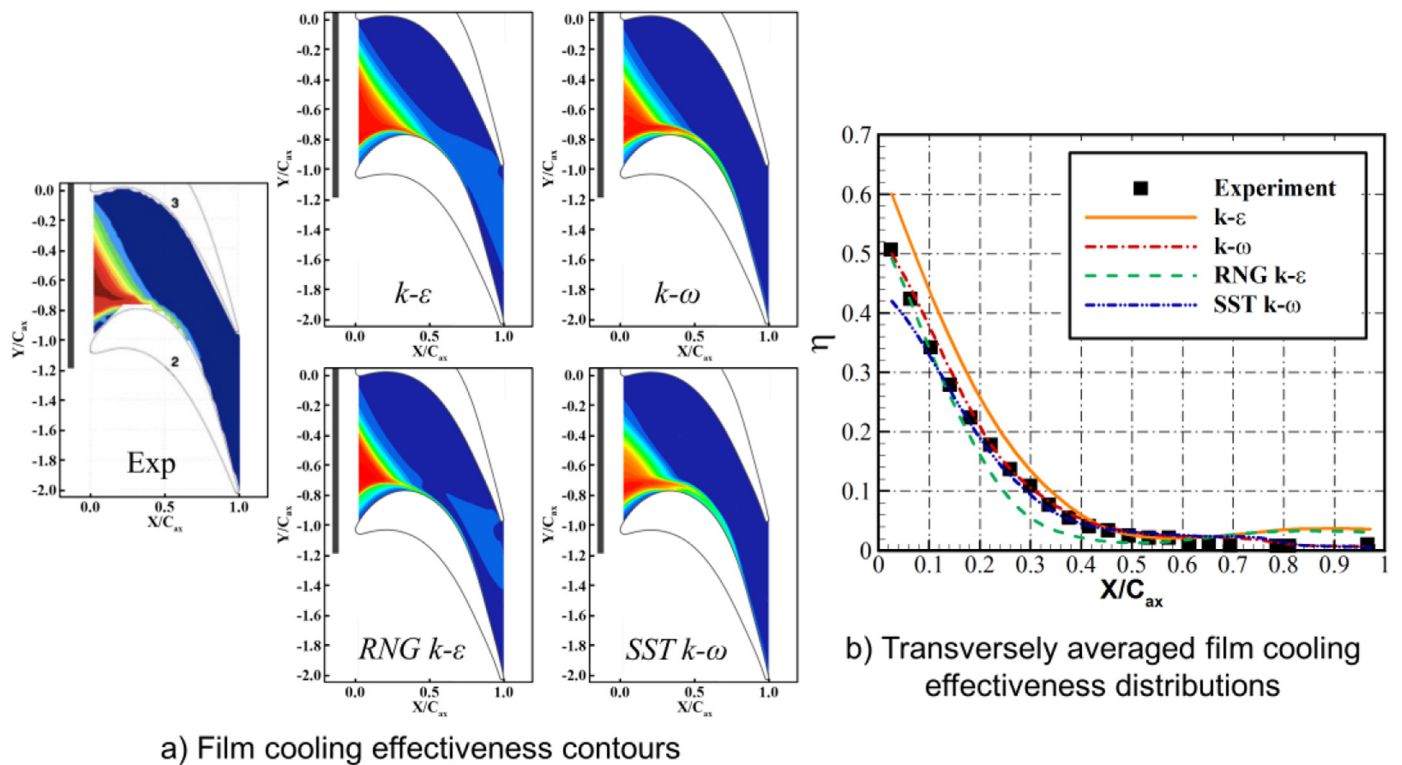


Fig. 4. Comparison of the measured and predicted endwall film cooling effectiveness.

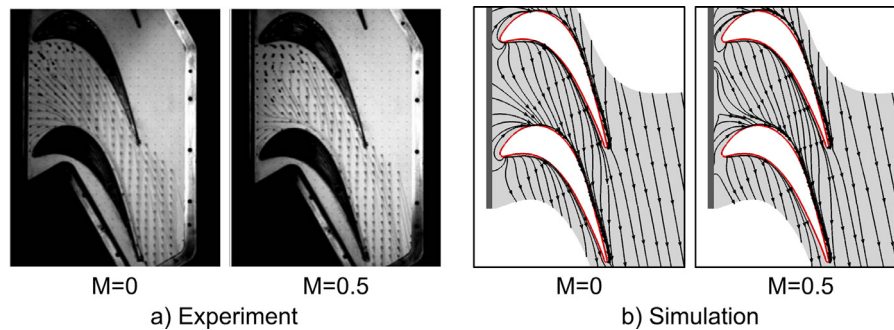


Fig. 5. Comparison of the near-endwall limited streamlines.

Fig. 5 further gives a comparison of the measured and predicted secondary flow patterns near the endwall. Fig. 5(a) and (b) show the endwall limited streamlines measured by oil-dot visualization in Ref. [2] and predicted by the standard  $k-\omega$  model, respectively. Results show that the selected  $k-\omega$  model accurately captures the shape, location and trajectory of the near-endwall secondary flows under different blowing ratios. It could be expected that a good capture of the secondary flows can ensure a reasonable prediction for the aerodynamic and thermal performance. Therefore, the standard  $k-\omega$  model is finally adopted due to its good prediction for both the film-cooling behaviors and secondary flow patterns.

### 3. Non-axisymmetric endwall design and optimization method

#### 3.1. Parameterization method for NEC

The horseshoe vortex and transverse cross flows are both the prominent secondary flow structures in a blade passage. The horseshoe vortex makes it challenging to cool the junction between endwalls and blade LE and is also considered as a primary source of the passage vortex. The transverse cross flows tend to strengthen the passage vortex and sweep the coolant towards

the blade suction side. Aiming to diminish the adverse impacts of these secondary flow structures, this paper proposes a novel non-axisymmetric endwall contouring approach. Unlike the traditional methods that only focus on the cross flows, this full-endwall contouring method has two purposes. Firstly, a ramp-like geometry is formed near the blade LE to weaken the horseshoe vortex. Secondly, a non-axisymmetric contoured endwall is also generated between the two adjacent blades to weaken the transverse cross flows.

Fig. 6 depicts the parameterization of the non-axisymmetric endwall contouring method. The contoured endwall is clearly divided into two parts. At the forepart of the endwall, the non-axisymmetric contouring is merely formed near the blade LE. At the rear part of the endwall, the non-axisymmetric contouring is formed between the two adjacent blades. The shape of the entire contoured endwall can be described by the combination of a tangential contouring curve and an axial amplitude curve. The control curves of the fore-part and the rear-part are colored in green and blue, respectively. As shown in Fig. 6, the tangential contouring curves of the fore part and rear part are different. For the fore part, the tangential contouring curve consists of two half-period trigonometric functions, one near the pressure side and the other

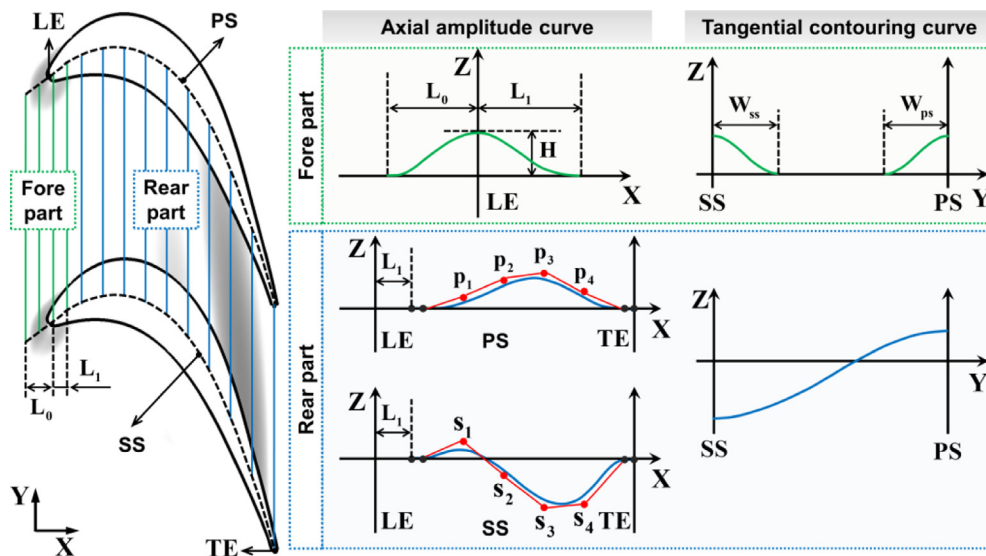


Fig. 6. The parameterization method for NEC.

near the suction side. The periods of the two trigonometric functions determine the contoured region's tangential width at each side ( $W_{ps}$  and  $W_{ss}$ ), respectively. The two tangential profiles reach their peaks at the blade camber line and their maximum heights are determined by the fore-part axial amplitude curve. Similarly, this axial amplitude curve is also formed by combining two half-period trigonometric functions to guarantee a smooth transition to the plain endwall. The maximum amplitude ( $H$ ), the starting ( $L_0$ ) and ending position ( $L_1$ ) of the contoured zone are controlled by the fore-part axial amplitude curve. In all, to generate the fore-part contoured endwall, five design parameters are needed in total.

For the rear part, the tangential profile curve starts from the camber line of the pressure-side blade (PS) and ends at the camber line of the suction-side blade (SS). To form a smooth concave/convex surface, the tangential profile is fitted by a half-period trigonometric function, whose period equals twice the blade pitch. The heights of the tangential profile at the PS and SS are controlled by the rear-part axial amplitude curves at the PS and SS, respectively. The rear-part axial amplitude curve is fitted by a non-uniform B-Spline, which ends at the blade trailing edge. To achieve a reasonable degree of design freedom, the axial amplitude curve is determined by eight control points in total. The middle four control points are allowed to be moved along the span-wise direction ( $Z$ -axis), while the first two and the last two control points are fixed for a smooth connection. Therefore, the shape of the rear-part contoured endwall can be modified by adjusting the  $Z$  coordinates of the control points at the PS side ( $P_1, P_2, P_3, P_4$ ) and the SS side ( $S_1, S_2, S_3, S_4$ ). Note that, a positive  $Z$  coordinate forms a convex surface, while a negative  $Z$  coordinate forms a concave surface. In all, a total of eight design parameters are needed to determine the rear-part contoured endwall. Taken together, the entire contoured endwall can be described with a total of 13 design parameters. Fig. 7 illustrates an example of a typical non-axisymmetric contoured endwall. The contoured endwall is constructed by lofting a series of tangential profile curves. For this case, it has two remarkable non-axisymmetric contouring features: a protrusion near the blade LE and a valley near the blade suction side.

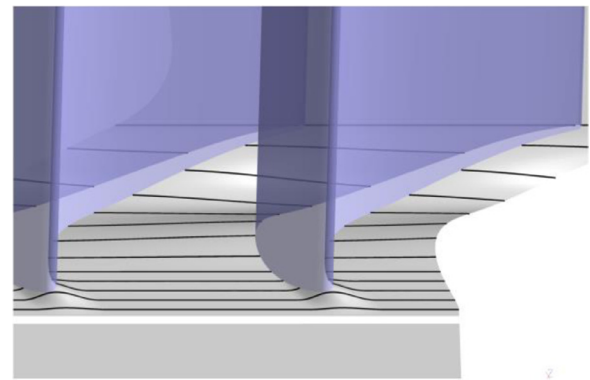


Fig. 7. The illustration of a typical NEC design.

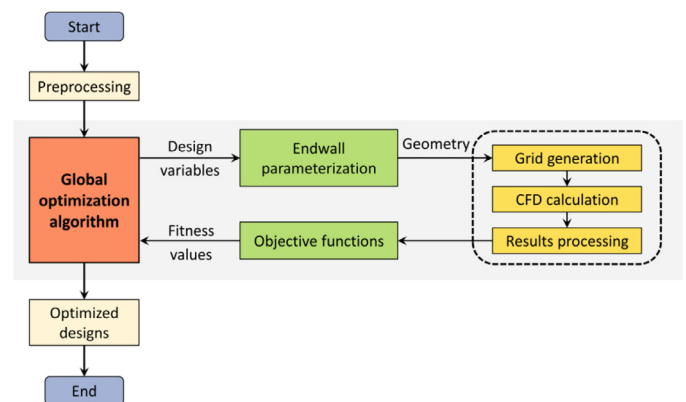


Fig. 8. The flowchart of aero-thermal optimization platform for NEC.

### 3.2. A brief introduction of the optimization platform for NEC

In this paper, optimization is used as a tool to extract representative optimal solutions in the NEC design space to reveal the influence mechanism of NEC on endwall aerodynamics and ther-

mal management. A brief introduction of the optimization platform for NEC is given below. Fig. 8 presents the flowchart of the aero-thermal optimization for a non-axisymmetric endwall. This flowchart combines the NEC parameterization method, CFD automatic computations and an efficient global optimization algorithm. The NEC parameterization method described above is used to generate the geometric shape of each individual in a given design space. In other words, the parameterization method converts the design variables into geometric entities for CFD calculations.

**Table 4**  
Optimization variables and their variation ranges.

| Parameters<br>Range | H<br>[0.005S, 0.025S] | $W_{ps}, W_{ss}$<br>[0.1P, 0.2P] | $P_1, P_2, P_3, P_4$<br>[-0.025S, 0.025S] | $S_1, S_2, S_3, S_4$<br>[-0.04S, 0.04S] |
|---------------------|-----------------------|----------------------------------|-------------------------------------------|-----------------------------------------|
|---------------------|-----------------------|----------------------------------|-------------------------------------------|-----------------------------------------|

In order to achieve an automatic optimization for such a computationally expensive problem, CFD calculations should be run in batches. The automatic templates for both the mesh generations, CFD calculations and post-processing are generated using the script recording techniques. The Kriging-based efficient global optimization (EGO) algorithm is used to control the optimization process and search for the required aero-thermal optimal solutions. The EGO algorithm is suitable for non-linear and computationally expensive optimization problems [36,37]. There are three steps involved in a common EGO algorithm: generation of space-filling samples, construction of a high-accuracy surrogate, and selection of infill criterions. More details about the mathematical function tests and engineering applications of the EGO algorithm can be found in Ref. [38].

### 3.3. Design variables and objective functions

As described in Section 3.1, there are 13 design parameters in total for the description of a NEC. During the optimization, only 11 design parameters, as listed in Table 4, are taken as optimization variables. The position parameters ( $L_0$  and  $L_1$ ) of the contoured region are maintained at reasonable values ( $L_0 = 0.1C_{ax}$ ,  $L_1 = 0.15C_{ax}$ ) and are not optimized. The variation ranges of each optimization variable are summarized in Table 4. Note that the maximum amplitude (H) of the fore-part NEC and the Z coordinates of the control points ( $P_i$ ,  $S_i$ ,  $i = 1 \sim 4$ ) of the rear-part NEC are normalized by the blade span (S). The tangential widths of the fore-part NEC ( $W_{ps}$  and  $W_{ss}$ ) are normalized by the blade pitch (P). For this 11-dimensional optimization problem, there is a total of 73 samples. 67 of them are initial samples, which are uniformly distributed in the design space. The remaining 6 samples are added based on the EI infill criterion.

It is well established that, the NEC has a significant influence on both the aerodynamic and thermal performance of a turbine blade endwall, and the influences on different performance indicators are probably different. In other words, there may be trade-offs between different performance indicators. Therefore, it is very essential to consider different performance indicators as comprehensively as possible in the design and optimization of the NEC. For this reason, three important performance indicators involved in the aero-thermal design of turbine endwalls, namely aerodynamic loss (AD), averaged film cooling effectiveness (FC) and averaged heat transfer coefficient (HT), are all taken as optimization objectives in this paper. In order to exclude the influence of the three-objective interactions and clarify the fundamental flow-thermal physics due to NEC as much as possible, three single-objective optimizations are separately performed to search for representative solutions. The optimization is performed at MFR=1.0%, which is a representative condition. Besides, there is no need to set a constraint for the mainstream mass flow because the velocity boundary condition is imposed at the mainstream entrance. Overall, the aero-thermal optimization problems in this paper can be defined as below:

$$\min F_1(\mathbf{x}) = \bar{\xi}_{p,t} \quad (1)$$

$$\max F_2(\mathbf{x}) = \eta_{ave} \quad (2)$$

$$\min F_3(\mathbf{x}) = (h/h_0)_{ave} \quad (3)$$

where  $F_1(\mathbf{x})$ ,  $F_2(\mathbf{x})$ ,  $F_3(\mathbf{x})$  denotes the three objective functions;  $\bar{\xi}_{p,t}$  represents the total pressure loss coefficient of the entire blade passage;  $\eta_{ave}$  denotes the area-averaged film cooling effectiveness of the first half portion. This is because that the purge coolant only covers the first half of the endwall, due to the entrainment of the passage vortex.  $(h/h_0)_{ave}$  denotes the area-averaged heat transfer coefficient of the entire endwall. Note that,  $h_0$  represents the heat transfer coefficient without the purge slot. The specific definitions of the formulas used in this paper are listed below:

$$\bar{\xi}_{p,t} = \frac{\frac{m_\infty}{m_\infty + m_c} p_{t,\infty} + \frac{m_c}{m_\infty + m_c} p_{t,c} - p_{t,out}}{\rho_\infty U_\infty^2 / 2} \quad (4)$$

$$\eta = (T_\infty - T_{aw}) / (T_\infty - T_c) \quad (5)$$

$$h = q_w / (T_{aw} - T_w) \quad (6)$$

$$Nu = hC / \lambda \quad (7)$$

$$MFR = m_c / m_\infty \quad (8)$$

where  $p_{t,c}$ ,  $p_{t,\infty}$  and  $p_{t,out}$  are the total pressure at the entrance of the slot, the entrance of the blade and the exit of the blade, respectively;  $m_c$  and  $m_\infty$  are the mass flow rates of the cooling air and the hot mainstream, respectively;  $T_c$  and  $T_\infty$  are the temperature of the cooling air and the hot mainstream, respectively;  $T_{aw}$  and  $T_w$  denote the adiabatic wall temperature and the non-adiabatic wall temperature, respectively;  $q_w$  represents the heat flux;  $\lambda$  represents the fluid's thermal conductivity.

During the optimization, three high-accuracy surrogates are firstly constructed for different performance indicators (AD, FC, and HT), respectively, using the Kriging model. Then, each surrogate is optimized separately using the EGO algorithm to find the corresponding optimal solution (AD-Opt, FC-Opt, and HT-Opt). Finally, the aero-thermal performance of the baseline and optimal designs are compared under different MFRs, and the underlying flow and thermal fundamentals behind the performance differences are elucidated.

## 4. Results and discussion

### 4.1. Optimization results

After the aero-thermal optimization, we obtain three optimal NEC designs with the best aerodynamic performance (AD-Opt), the best film cooling performance (FC-Opt), and the best heat transfer performance (HT-Opt). Fig. 9 compares the geometry of the three optimal NEC designs with the baseline flat endwall. Fig. 9(a) compares the geometric profiles and Fig. 9(b) compares the amplitude distributions. It is noteworthy that all three optimal designs have a ramp-shaped profile in front of the blade LE. This suggests that a ramp-shaped profile designed to weaken the horse-shoe vortex is beneficial for both aerodynamic and thermal performance. The AD-Opt and the HT-Opt are similar in shape, with a slight protrusion near the fore-part of the blade suction side (SS for short) and a distinct valley at the rear-part endwall, especially near the blade SS. The difference is that the AD-Opt is more concave near the rear-part of the blade SS. In contrast, FC-Opt has a significant convex profile at the middle part of the blade

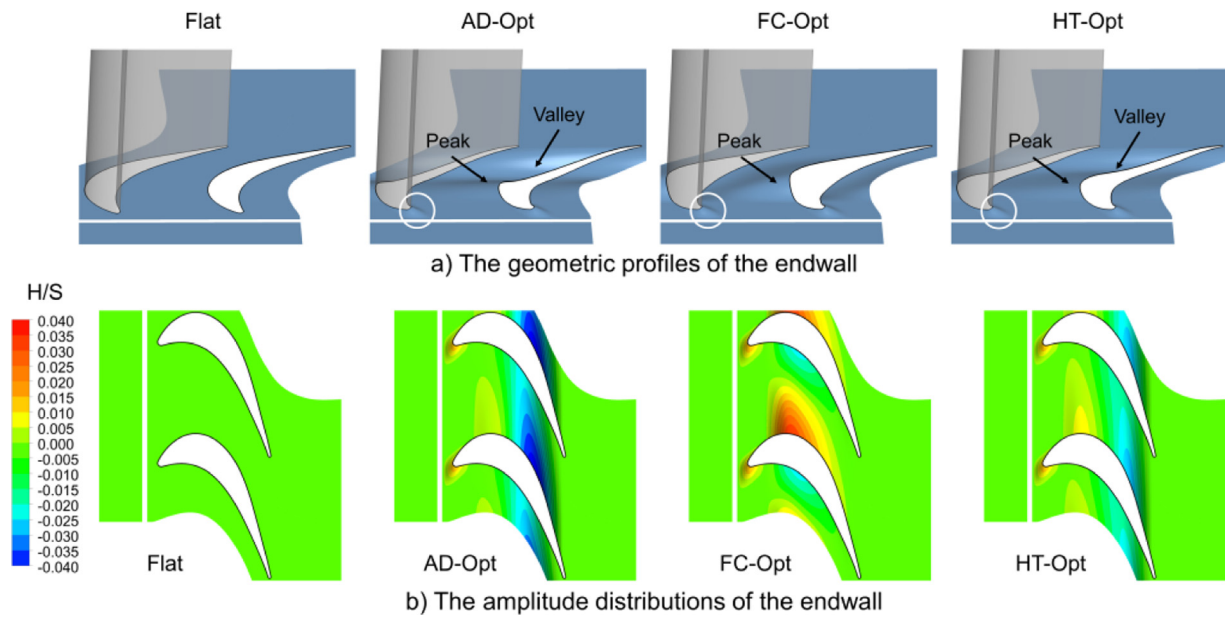


Fig. 9. Geometric comparison between the optimal NEC designs and the flat endwall.

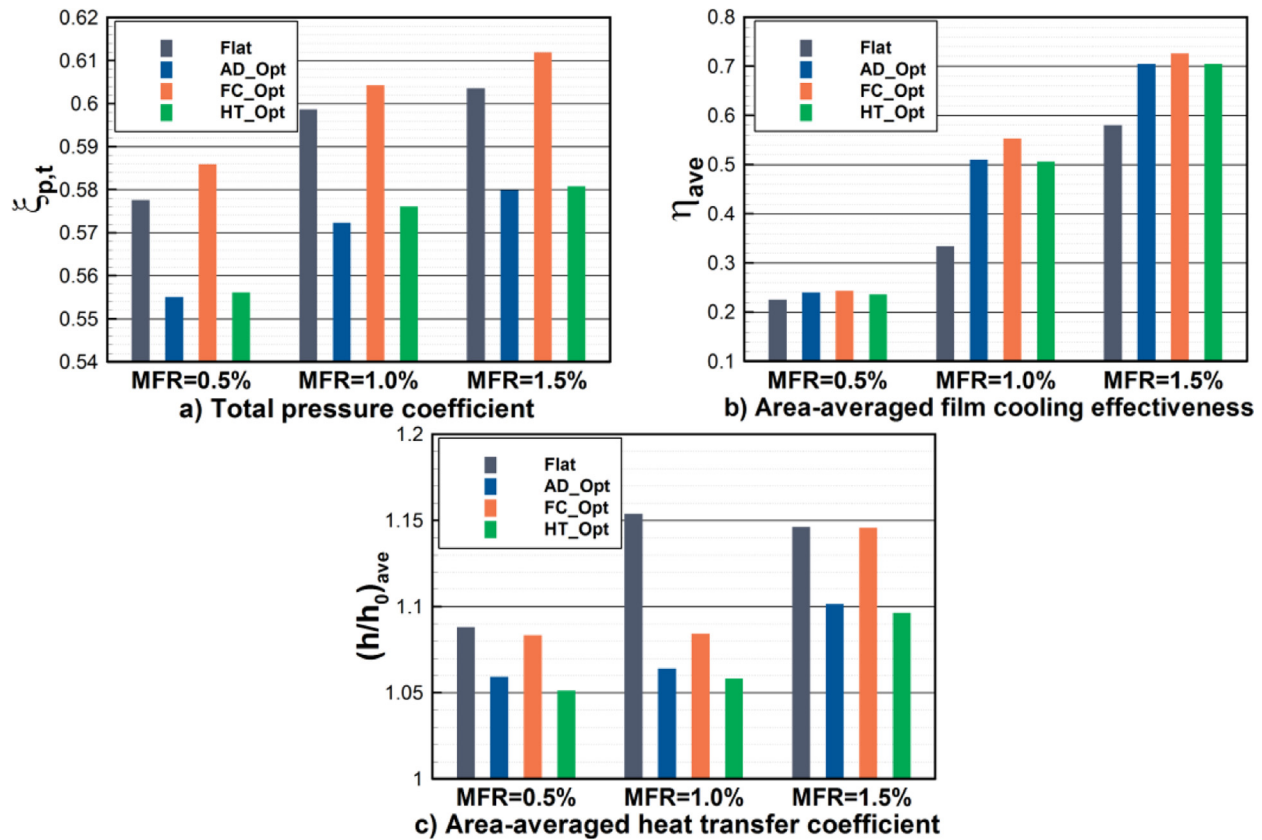


Fig. 10. Performance comparisons of optimal NEC designs and a flat endwall.

SS, without any concave profile near the rear-part of the blade SS. In general, the non-axisymmetric profile of the optimal NEC designs is mainly manifested near the blade LE and the blade SS, while it is not as pronounced near the blade pressure side (PS for short). Overall, the AD-Opt and the HT-Opt have some consistency, while there is a clear trade-off between the FC-Opt and them.

Fig. 10 presents quantifiable comparisons of total pressure loss coefficient (TPLC), area-averaged film cooling effectiveness (FCE) and area-averaged heat transfer coefficient (HTC) for three typical optimal designs and the flat endwall under different MFRs. As shown in Fig. 10(a), the AD-Opt achieves the best aerodynamic performance under all the MFRs investigated, closely followed by the HT-Opt. Both designs can reduce the TPLC significantly and to a comparable extent. In quantitative terms, the AD-Opt can de-



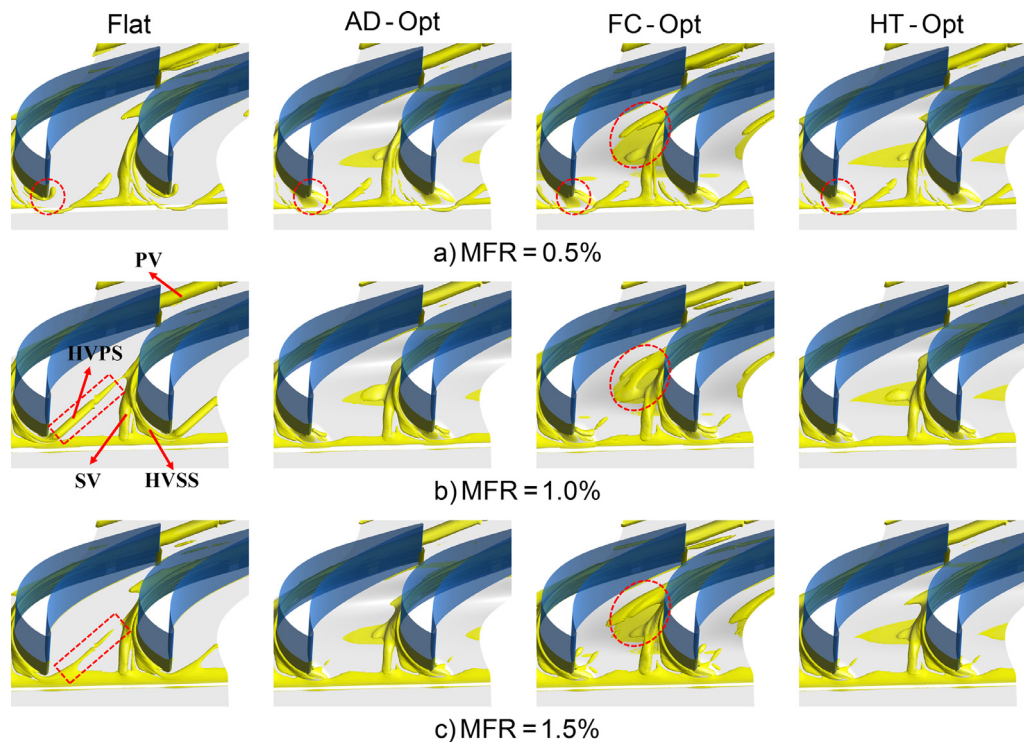


Fig. 11. Visualization of secondary vortex structures based on Q-criteria.

crease the TPLC by 3.9%, 4.4%, and 4.0% at MFRs of 0.5%, 1.0%, and 1.5%, respectively. In contrast, the FC-Opt increases the TPLC instead, albeit by a small amount. It is clear from Fig. 10(b) that all the three optimal designs can effectively enhance the area-averaged FCE, especially when  $MFR > 1.0\%$ . As expected, the FC-Opt performed the best, with the AD-Opt and the HT-Opt closely following. Quantitatively, the FC-Opt can enhance the area-averaged FCE by 8.3%, 65.2%, and 25.1% at MFRs of 0.5%, 1.0%, and 1.5%, respectively. It is worth noting that the improvements of film cooling by AD-Opt and HT-Opt are also quite considerable. The improvements can reach 50%, 20% at MFRs of 1.0%, and 1.5%, respectively. In regards to heat transfer performance, the HT-Opt achieves the lowest area-averaged HTC, followed closely by the AD-Opt, as shown in Fig. 10(c). Quantitatively, the HT-Opt can decrease the area-averaged HTC by 3.4%, 8.3%, and 4.4% at MFRs of 0.5%, 1.0%, and 1.5%, respectively. In contrast, the FC-Opt can only reduce the area-averaged HTC at  $MFR = 1.0\%$ , without any noticeable effect on HTC at  $MFR = 0.5\%$  and 1.5%. Overall, the FC-Opt achieves a quite significant improvement in film cooling, however, it does not perform as well in terms of aerodynamics and heat transfer. The AD-Opt and HT-Opt are comparable in all aspects of the aero-thermal performance, due to their similar geometries. They bring significant and relatively balanced improvements in aerodynamics, film cooling and heat transfer performance. The above results strongly confirm the effectiveness of the proposed NEC design and optimization method. In order to adequately reveal the underlying flow and heat transfer fundamentals of the NEC designs and therefore to support the above optimization results, detailed and in-depth flow and thermal field analyses are performed in the following sections.

#### 4.2. Flow field and aerodynamic analysis

The secondary flow structures and the interactions between the coolant jets and hot mainstream fundamentally determine the aerodynamics performance and thermal management of blade endwalls. It is therefore essential to elucidate the effect of NEC on

the flow and thermal fields near the endwall. Fig. 11 provides an insight into the near-endwall secondary vortex structures with the help of vortex core trajectory visualizations. Fig. 11(b) marks the location and migration trajectory of the primary secondary vortices. The horseshoe vortex (HV for short) originates at the juncture of the blade LE and the endwall, where the inlet boundary layer strikes the blade LE and rolls up. The HV is subsequently developed into two branches with opposite rotation directions, with its PS branch (HVPS for short) crossing the blade channel and its SS branch (HVSS for short) moving closely along the SS surface of the blade. Another noticeable vortex is the passage vortex (PV for short), which is fed and energized by the strong cross flows from PS to SS. It is clearly observed that the HVPS is eventually entrained into the PV, further intensifying the PV. In addition, there is a separation vortex (SV for short) downstream of the purge slot, where the injected coolant firstly separates and then reattaches.

The optimal NEC designs modify the secondary vortex structures in two primary ways. Firstly, all three NEC designs can substantially diminish the size and strength of the HV and its two branches. Especially when  $MFR = 1.0\%$ , the HVPS is weakened very significantly. This is attributed to the ramp-like profile near the blade LE, which is designed to weaken the HV. On the other hand, the rear-part profile is intended to control the secondary flow patterns by accelerating or decelerating the local fluid to alter the local pressure distribution. The AD-Opt and HT-Opt with concave profiles near the rear of the blade SS help to diminish the cross-flows and thus make the PV much weaker. However, the FC-Opt instead intensifies the cross flows there and promotes the development and growth of the PV. A more quantitative analysis of the impact of the NEC designs on the HV and the PV will be provided below.

Fig. 12 illustrates the formation and development of the HV using a cluster of streamlines facing the blade LE and several cross-sections colored by the turbulence kinetic energy. It is noted that a set of cases without purge flow is also included to better eluci-

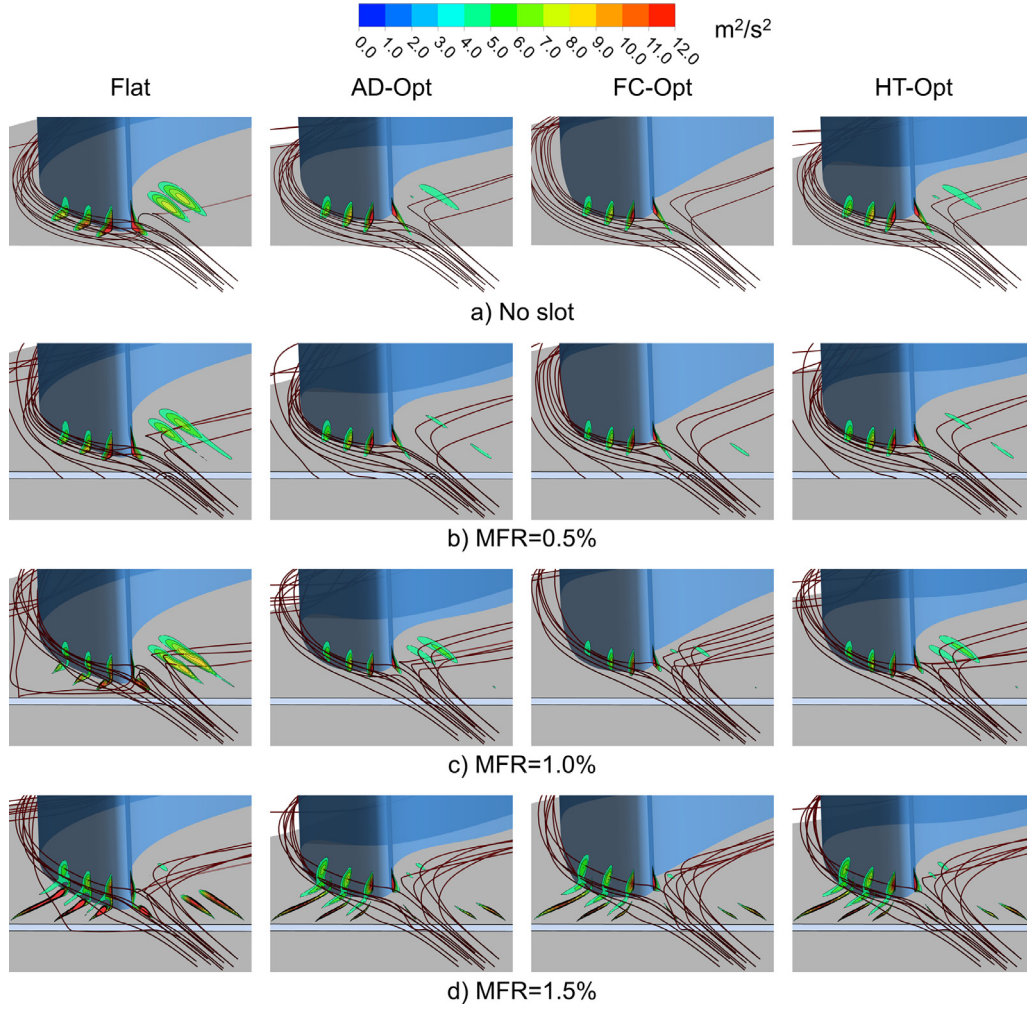


Fig. 12. The formation and development of the horseshoe vortex.

date the influences of endwall contouring. The oncoming boundary layer fluid impinges on the blade LE and stagnates, leading to reverse pressure gradients along both the streamwise and radial directions. The reverse pressure gradients make the near-endwall stagnant fluid roll up and flow backward, eventually forming the horseshoe vortex. Fig. 12 clearly illustrates that the optimal NEC designs result in a significant reduction in the size of the high turbulent kinetic energy region caused by the HV. It indicates that the NEC designs with a slope-like profile near the blade LE can significantly weaken the HV and its two branches compared to a flat endwall. There are two reasons for this observation. Firstly, the acceleration of oncoming fluids through a gradually rising slope helps to resist the inverse pressure gradient caused by the mainstream stagnation, thus suppressing the HV's formation and development. In addition, the smooth transition from the endwall to the blade LE eliminates sharp corners at the junction, enabling the boundary layer fluid to move along the endwall surface nicely, thus effectively inhibiting the backflow.

Our proposed NEC method also aims to weaken the strong cross flows caused by the cross-passage pressure gradient. Fig. 13 compares the static pressure coefficient ( $C_p$ ) distributions along the blade surface at the 5% span. The  $C_p$  is defined as below:

$$C_p = \frac{p - p_{out}}{p_{t,\infty} - p_{out}} \quad (9)$$

where  $p_{t,\infty}$  denotes the total pressure of the mainstream entrance;  $p_{out}$  represents the static pressure of the mainstream exit;  $p$  represents the static pressure at the blade surface.

Fig. 13(a) clearly illustrates the pressure difference between the blade PS and SS using the static pressure coefficient. Fig. 13(b) shows schematically the cross-flow patterns and the location where the pressure is extracted. It is well known that the lateral pressure difference contributes directly to the development of the cross flows, which ultimately impacts the strength of the PV. The basic idea of NEC is to generate an undulating surface to accelerate or decelerate the local fluid, thus altering the lateral pressure difference. The three NEC designs significantly modify the static pressure of the SS and marginally modify that of the PS compared to the flat endwall. The AD-Opt and the HT-Opt which have distinct valleys near the middle and rear of the SS decelerate the flow locally and therefore increase the static pressure of the region ( $0.45 < X/C_{ax} < 0.8$ ). The FC-Opt, however, has a significant protrusion near the middle of the SS, which accelerates the local flow and therefore lowers the static pressure of the region ( $0.3 < X/C_{ax} < 0.65$ ). As a result, the AD-Opt and the HT-Opt can reduce the lateral pressure difference to inhibit the cross flows, while the FC-Opt increases the lateral pressure difference to promote the cross flows.

Fig. 14 plots the TPLC distributions at 20%  $C_{ax}$  behind the trailing edge of the blade. Two noticeable high-loss areas are identified

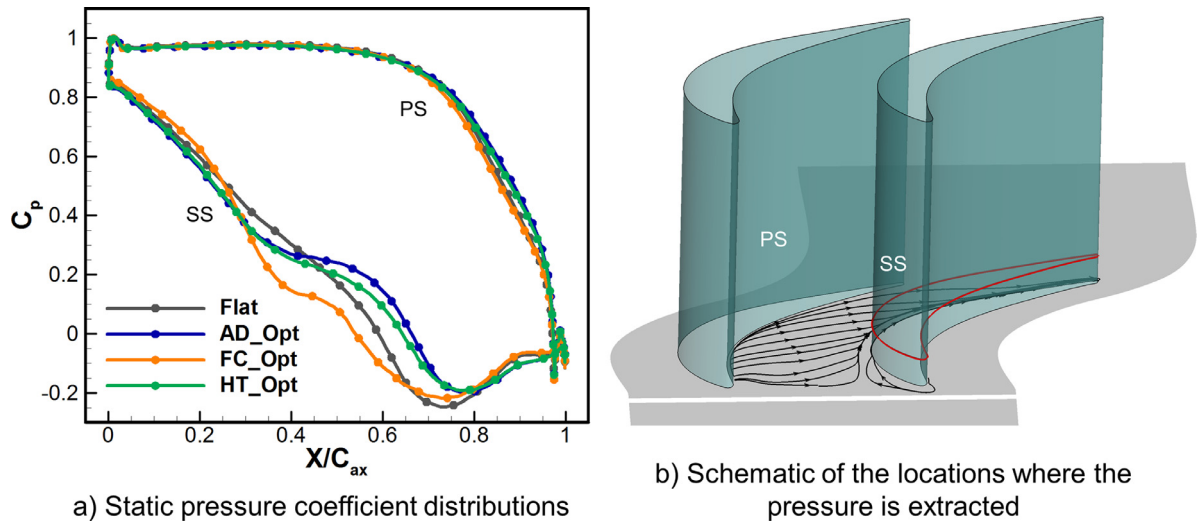


Fig. 13. The static pressure coefficient distributions along the blade surface at a 5% span.

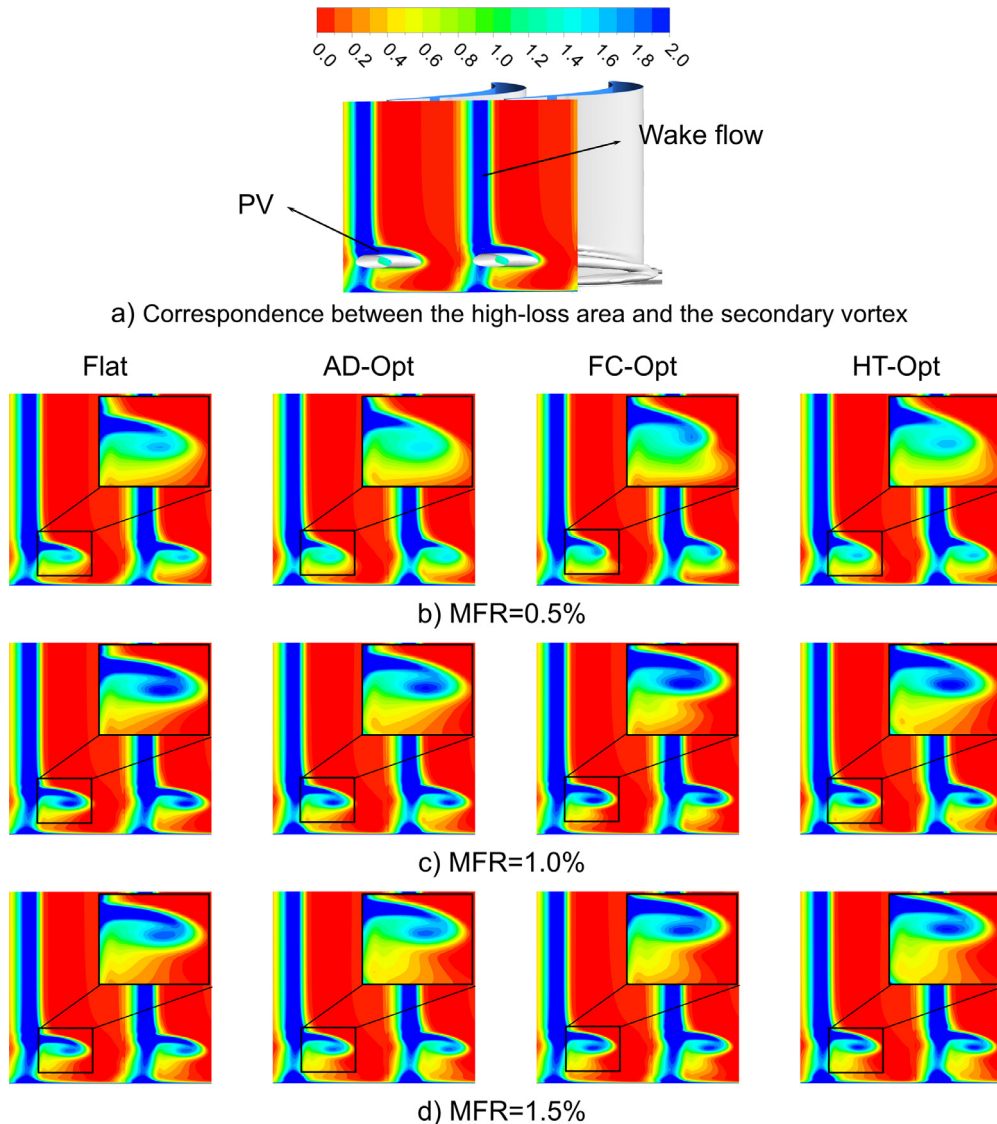


Fig. 14. Distributions of the total pressure loss coefficient.



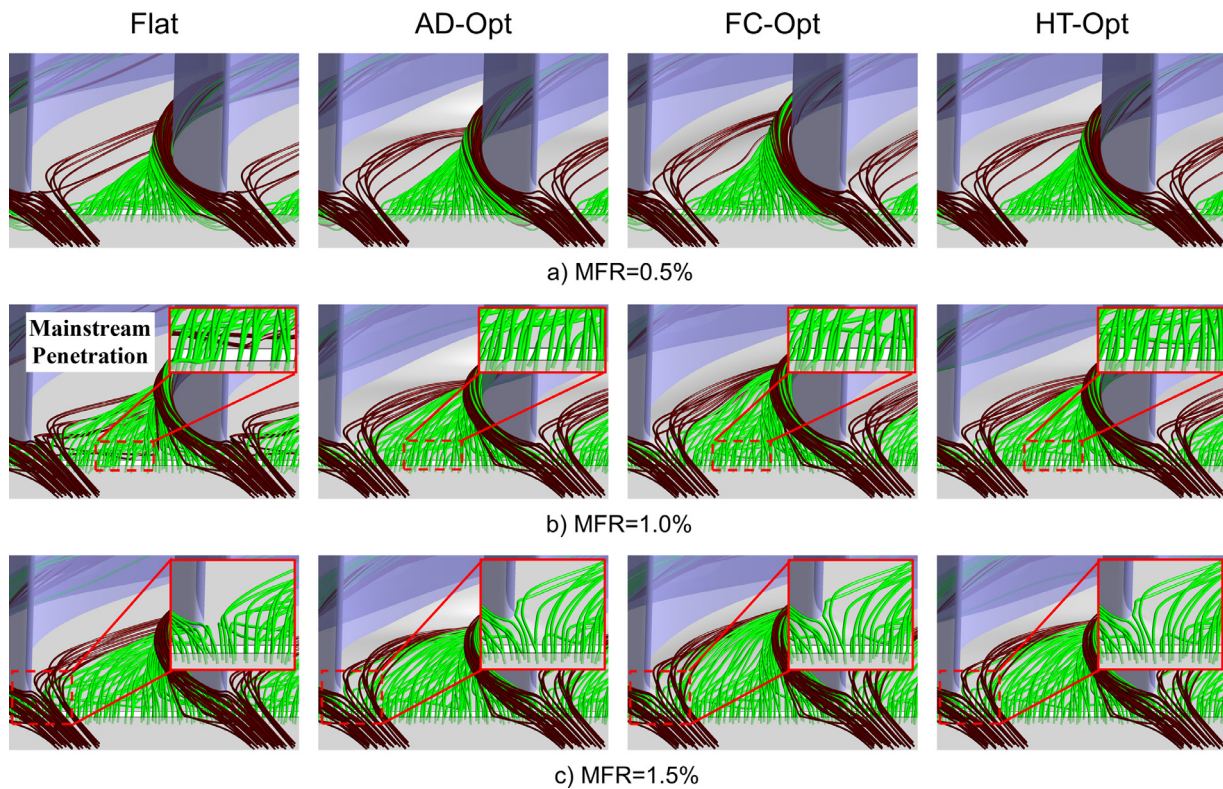


Fig. 15. Interactions between the mainstream flow and purge flow.

in Fig. 14(a), where the strip-shaped high-loss areas are associated with the blade wake flow and the oval-shaped high-loss areas are associated with the PV. It is apparent that the NEC designs primarily alter the PV intensity, so the high-loss regions related to the PV are magnified for clear comparisons. It is observed that the effects of NEC designs on the aerodynamic loss in the region associated with the PV are similar for different MFRs. The AD-Opt and the HT-Opt are shown to be effective in reducing the aerodynamic loss associated with the PV because of the reduced intensity of the PV. On the contrary, since the FC-Opt slightly intensifies the PV, it increases the loss there instead. This finding is consistent with the vortex analysis based on the Q criterion in Fig. 11. It is also worth noting that in addition to the strength change, the NEC designs also tend to elevate the passage vortex from the endwall.

#### 4.3. Film cooling performance analysis

In this section, the effects of optimal NEC designs on endwall film cooling characteristics are investigated and discussed. More importantly, this section also reveals the underlying fundamentals of how the NEC affects the cooling behaviors by altering secondary vortices and coolant-mainstream interactions.

The interactions between the purge coolant and the hot mainstream are important for the purge cooling behaviors of a turbine endwall. Fig. 15 plots the streamlines of the coolant and the mainstream, clearly revealing the inter-mixing and transport between them. The incoming mainstream flow is colored in brown, while the coolant discharged from the purge slot is colored in green. In general, the purge coolant forms a V-shaped coverage on the fore-part of endwalls and appears to be restricted to the blade SS by the HVPS. It is also noted that in the downstream region, the purge coolant is completely entrained in the PV and is almost blown away from the endwall, resulting in poor cooling at the rear of the endwall. This covering shape directly determines the FCE contours on blade endwalls. Apparently, a higher MFR can provide

much greater film-cooling coverage. The NEC designs show dramatic changes in the migration of coolant and its mixing with the mainstream, especially for MFE > 1.0%. Firstly, for the region near the blade LE, since all the three NEC designs can diminish the HV, the purge jet can spread closer to the blade LE and provide better cooling there. Moreover, thanks to the weakened HVPS, all the three NEC designs can prevent the mainstream from penetrating into the coolant, thereby diminishing the turbulent mixing and improving the local cooling effectiveness, especially for MFR = 1.0%. It is also noteworthy that the FC-Opt has broader film-cooling coverage than the other two NEC designs. This is because the coolant near the boundary layer is accelerated while passing through the convex profile and is thereby more resistant to the cross flows there, thus spreading farther in the lateral direction.

Fig. 16 compares the film cooling effectiveness (FCE) contours of the optimal NECs with the flat endwall. The FCE is distributed in a V-shape at the fore-part of the endwall, with better cooling near the SS and worse cooling near the PS. This observation is consistent with the distribution of streamlines in Fig. 15. The influence of NEC on endwall film cooling varies with MFRs. When MFR=0.5%, as shown by the white line in Fig. 16(a), there is a great lack of coolant coverage around the blade LE. Even with the NEC designs, it fails to achieve noticeable film-cooling enhancement for this low MFR. This is attributed to the occurrence of mainstream intrusion, which will be described in Fig. 17. Compared to the flat endwall, the FC-Opt can moderately improve the endwall FCE near the fore-part of the blade SS, where a convex profile is formed.

When the MFR is relatively high (MFR > 1.0%), the film cooling advantages of the optimal NEC designs are very considerable. When MFR = 1.0%, the NEC designs improve the FCE in three aspects. Firstly, the FCE value of the poorly-cooled region is significantly increased by the three NEC designs due to the weakening of turbulent mixing there. As a result, a more uniform FCE distribution is obtained, which is promising to reduce the thermal



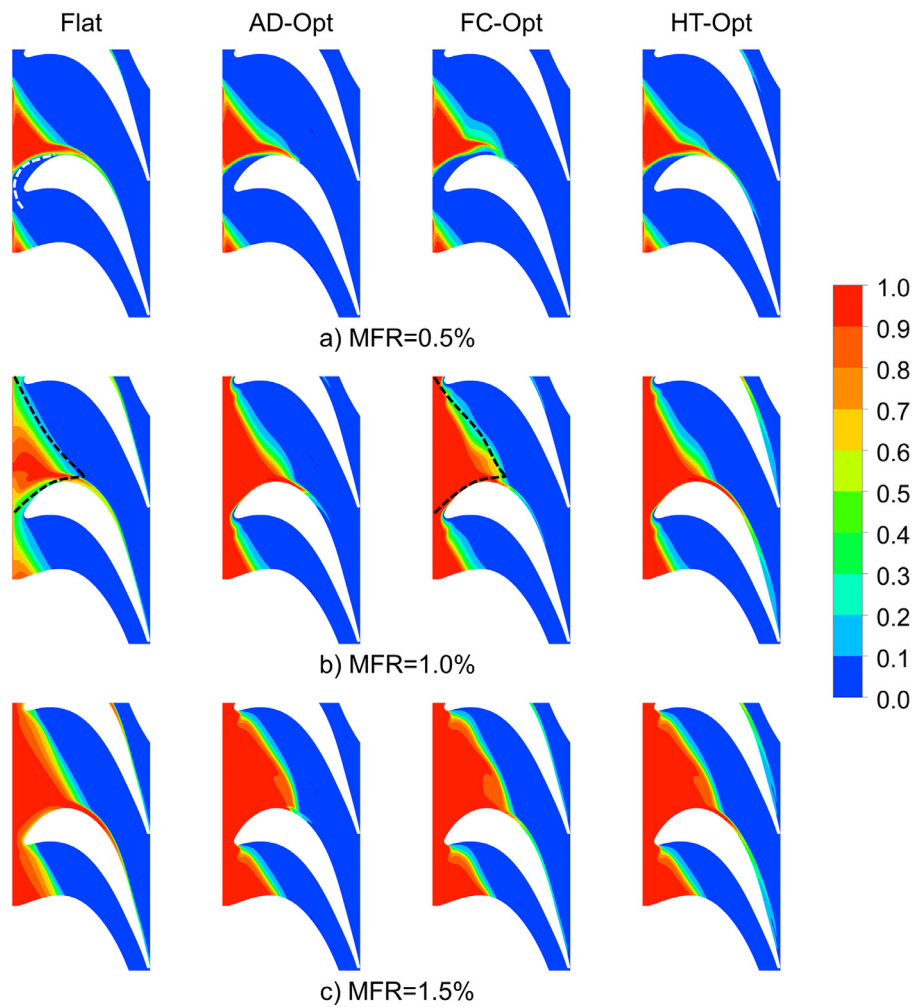


Fig. 16. Film cooling effectiveness contours comparison.

stress. Secondly, the film-cooling coverage for the endwall is also extended by the three NEC designs. Especially for the FC-Opt design, it dramatically enlarges the coverage area in the lateral direction as shown in Fig. 16(b). Finally, the three NEC designs also enhance the cooling efficiency for the junction between the blade LE and the endwall, where thermal management is challenging. When the MFR reaches 1.5%, the optimal NEC designs still enhance film cooling performance, although the improvement is not as significant as at MFR = 1.0%. The most noticeable improvement from the NEC designs at this MFR is the significant expansion of film-cooling coverage.

To better illustrate the flow and temperature field near the blade LE, and to exhibit the mainstream intrusion phenomenon at a low MFR, the streamlines and dimensionless temperature distributions on a cross-section vertical to the blade LE are plotted in Fig. 17. The definition of the dimensionless temperature ( $\varphi$ ) is given below:

$$\varphi = (T_{\infty} - T_f) / (T_{\infty} - T_c) \quad (10)$$

Where  $T_f$  represents the static temperature of the fluid at a certain location.

It is clearly seen from Fig. 17(a) that the hot main-flow intrudes into the inside of the purge slot at a low MFR (MFR=0.5%). In fact, the static pressure distribution of the mainstream near the slot exit is uneven, and the pressure near the stagnation point is higher than elsewhere. For this reason, there is almost no purge coolant injected into the mainstream near the blade LE at a very

low MFR. As a result, the purge flow can barely cover a small portion of the endwall, so the NEC design has a limited influence on film cooling behaviors at an MFR of 0.5%. When the MFR is greater than 1.0%, the NEC designs significantly alter the flow and thermal field near the blade LE. All three NEC designs have a rising slope in front of the blade LE. On the one hand, this slope weakens the HV and therefore reduces the turbulent mixing of the hot mainstream and coolant there. On the other hand, this rising slope allows the purge coolant to approach the blade LE and adhere closely to the endwall surface. In summary, the optimal NEC designs significantly enhance the cooling effectiveness around the blade LE compared to a flat endwall when MFR > 1.0%.

Fig. 18 displays the dimensionless temperature distribution over a series of cross-sections perpendicular to the axial direction. It is recognized that the spread and penetration of coolant into the hot mainstream largely determine the local cooling effectiveness. It is clearly visible that the complicated secondary vortices near the endwall heavily impact the spreading of the purge coolant and its mixing with the mainstream. The HV and its two legs tend to reinforce the turbulent mixing locally, while the cross flows tend to restrict the lateral spreading of the purge coolant. The mainstream ingestion that appears at MFR=0.5% also impacts the lateral transport of the purge coolant. Therefore, the NEC designs are intended to improve the temperature field distribution by altering the secondary vortices. All the three optimal NEC designs can weaken the HV, therefore reducing the turbulent mixing of regions

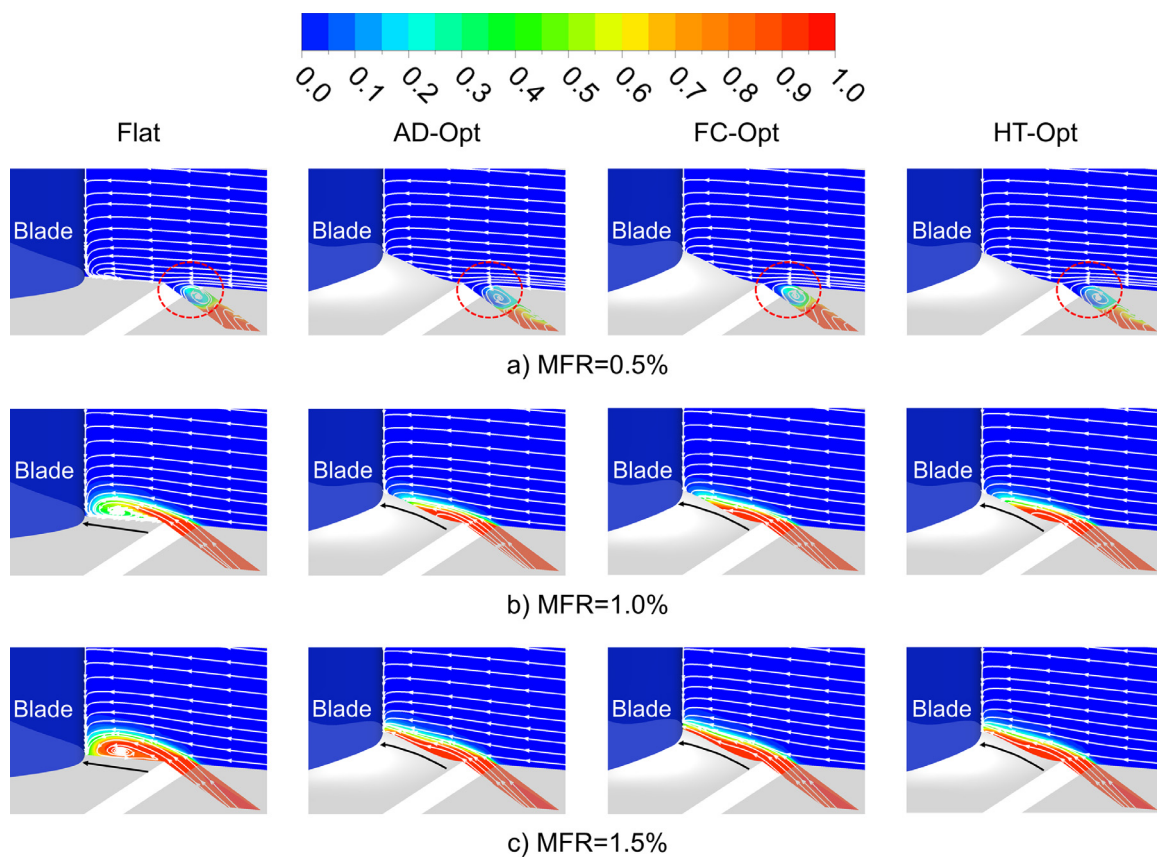


Fig. 17. Dimensionless temperature distributions and streamlines.

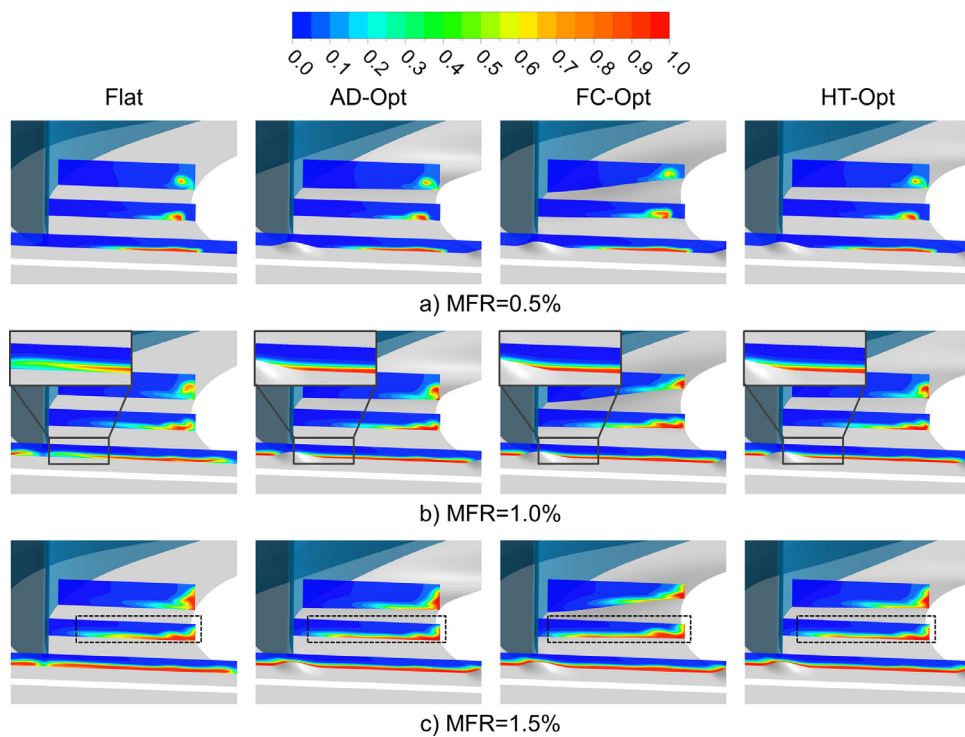


Fig. 18. Dimensionless temperature distributions over a series of cross-sections.

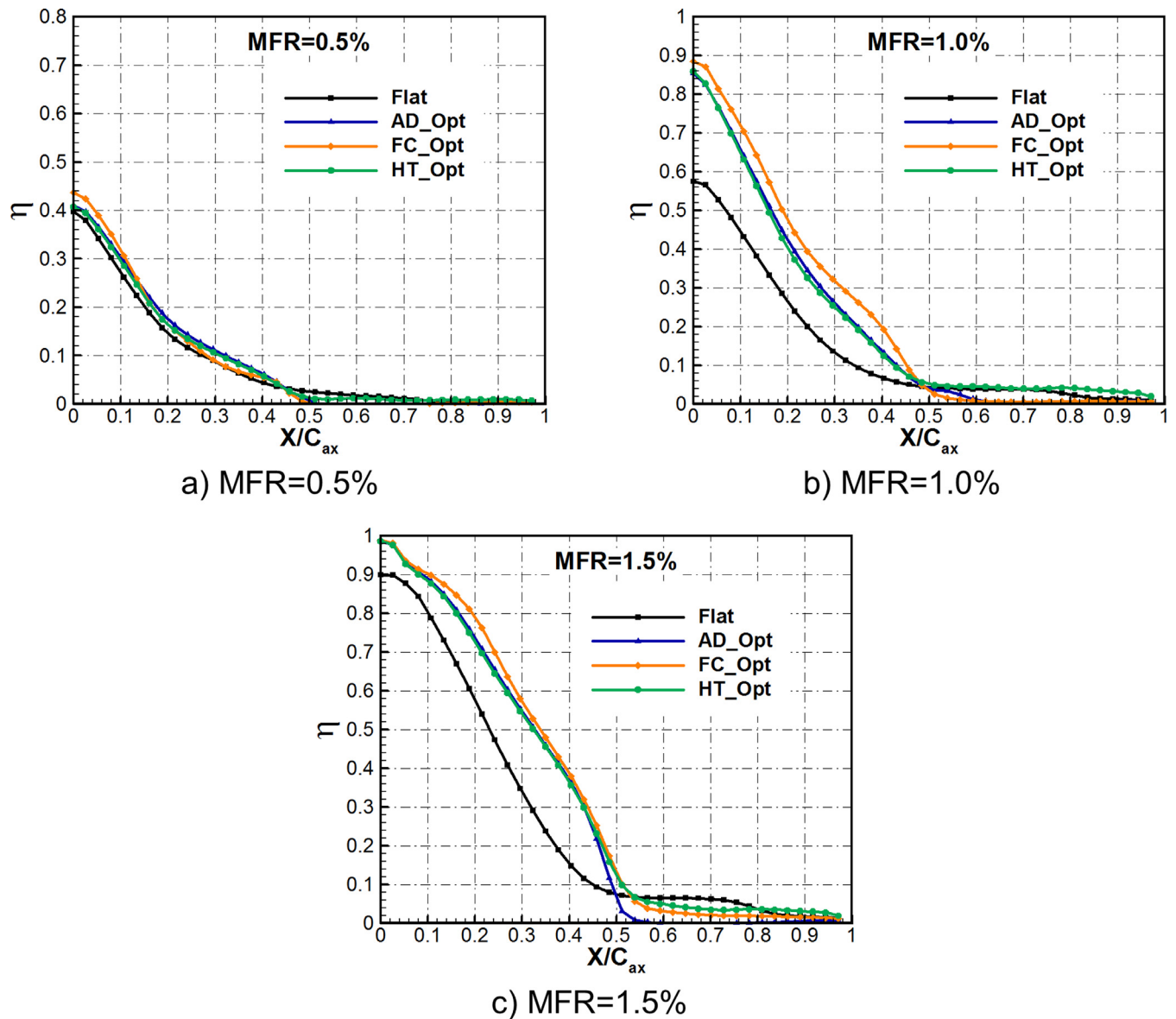


Fig. 19. Axial distributions of the laterally averaged FCE.

affected by the HV and HVPS. As shown in the magnified view of Fig. 18(b), the three NEC designs substantially diminish the penetration of the hot mainstream into the coolant film, resulting in a much cooler and more uniform temperature field there. Moreover, the three NEC designs enable the coolant ejection to spread much farther laterally and also stay closer to the endwall surface due to the weakened HVPS. In addition to the above benefits, the FC-Opt can further enlarge the lateral coverage since it accelerates the near-wall coolant to be more resistant to cross-flow effects.

Fig. 19 presents a quantified comparison for the laterally-averaged FCE along the axial location. The horizontal axes denote the normalized axial position coordinates and the vertical axes represent the laterally-averaged FCE. When MFR=0.5%, the optimal NEC designs can result in a small increase in the laterally-averaged FCE in comparison to a flat endwall. When the MFR is greater than 1.0%, the three optimal NEC designs yield a dramatic enhancement in the laterally-averaged FCE of the first half endwall ( $X/C_{ax} < 0.5$ ). It is true that the FC-Opt achieves the highest laterally-averaged FCE among the three optimal NEC designs. It is also encouraging to note that the optimized designs targeting aerodynamics and heat

transfer can still achieve considerable improvements in film cooling. Because of the similar geometries, the AD-Opt and HT-Opt provide roughly comparable enhancements to the laterally-averaged FCE. It is also noteworthy that the trends of film cooling improvement by the NEC designs vary with MFR. When MFR=1.0%, there is a greater film cooling improvement in more upstream zones. When MFR=1.5%, however, the greatest improvement appears at the middle of the endwall ( $X/C_{ax}=0.4$ ), rather than the fore-part of the endwall.

#### 4.4. Heat transfer performance analysis

In this section, the influences of the NEC designs on endwall heat transfer characteristics will be discussed firstly. Then, considering the results of film cooling and heat transfer together, the impact of the NEC designs on the total thermal load (net heat flux reduction) of the endwalls will be evaluated.

Fig. 20 compares the  $Nu$  distributions on the endwall. In order to provide a deep insight into the endwall heat transfer characteristics, three regions with relatively high Nusselt number are circled. The HV leads to a noticeable horseshoe-shaped high heat



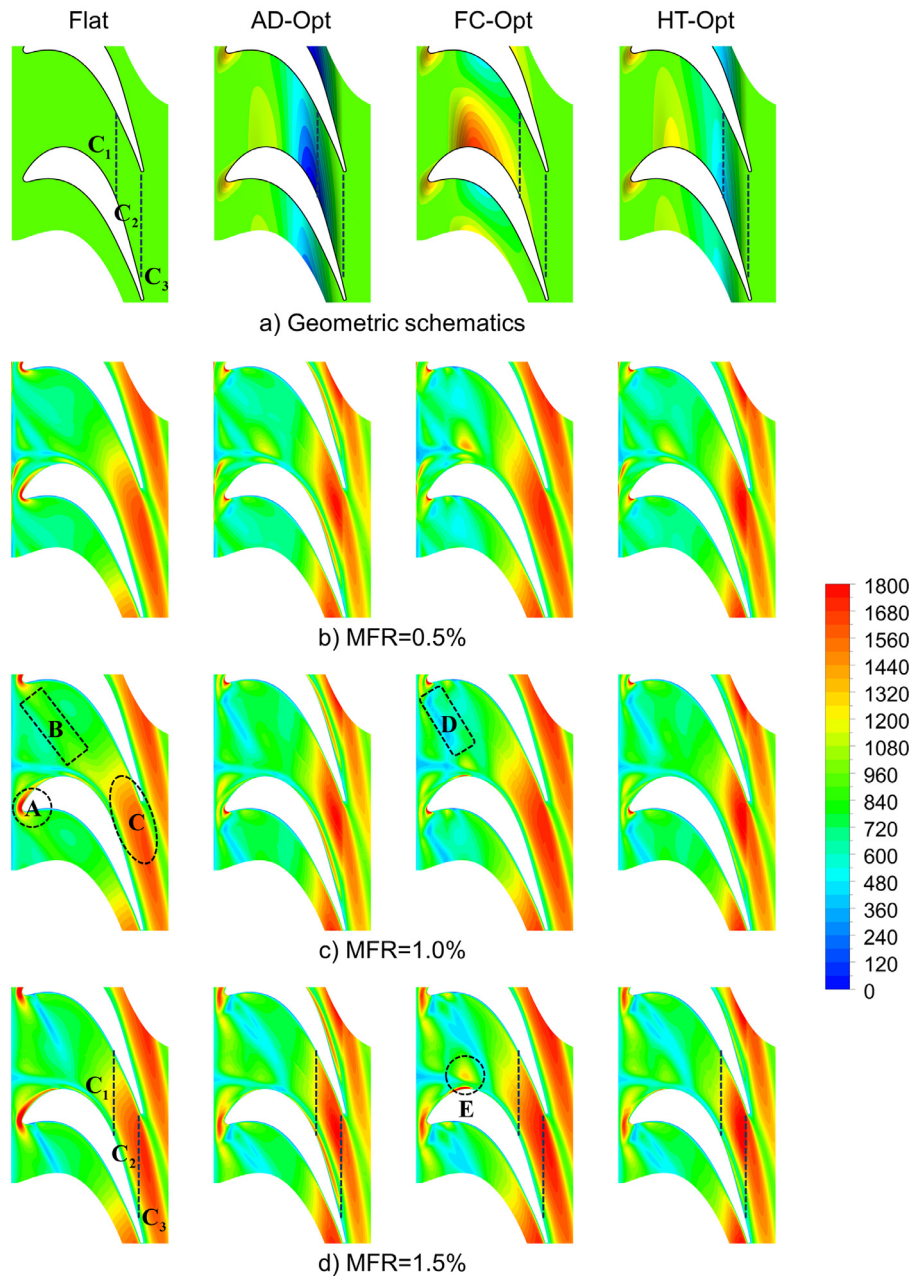


Fig. 20. Nusselt number contours on the endwall.

transfer zone A around the blade LE. The strong turbulent mixing and down-washing flows are responsible for the high  $Nu$  there. The transport of the HVPS from the blade LE to the SS of the adjacent blade forms a distinct high heat transfer trajectory B. Around the blade throat, there is also a very conspicuous high heat transfer zone C, which is caused by the flow acceleration and the PV. Another strip-shaped zone D with quite low  $Nu$  is also of interest. The occurrence of this region is largely attributed to the decrease in the turbulence intensity there.

All three NEC designs can significantly reduce the heat transfer intensity in zone A when compared to a flat endwall. When  $MFR < 1.0\%$ , the area of zone A is dramatically narrowed, and only a very small area with high heat transfer levels remains near the stagnation point. This is primarily attributed to the weakening of the HV near the blade LE. Accordingly, the high heat transfer trajectory B caused by the HVPS almost disappears for the NEC designs. It is worth noting that the effect of different NEC designs on

the heat transfer intensity of zone C varies considerably. Therefore, the high heat transfer zone C is divided into three subzones for a better interpretation. Compared to the flat endwall, the AD-Opt and the HT-Opt lower the heat transfer in zone  $C_1$  while augmenting that in zone  $C_2$ . This is because the concave profile changes the near-wall flow velocity. The gradually falling profile at zone  $C_1$  decelerates the flow locally and thereby reduces the heat transfer there, while the gradually rising profile at zone  $C_2$  accelerates the flow locally and thus intensifies the heat transfer there. Moreover, the AD-Opt and the HT-Opt are also shown to decrease the heat transfer in zone  $C_3$ . This can be attributed to the uplift of the PV as well as the reduction of PV intensity as illustrated in Fig. 14. In contrast, the FC-Opt enhances the heat transfer in both zone  $C_1$  and zone  $C_2$ . There are two possible reasons for this finding. On the one hand, the enhanced PV will intensify the local turbulent mixing. On the other hand, the upward protruding profile makes the PV closer to the endwall surface, further enhancing the heat

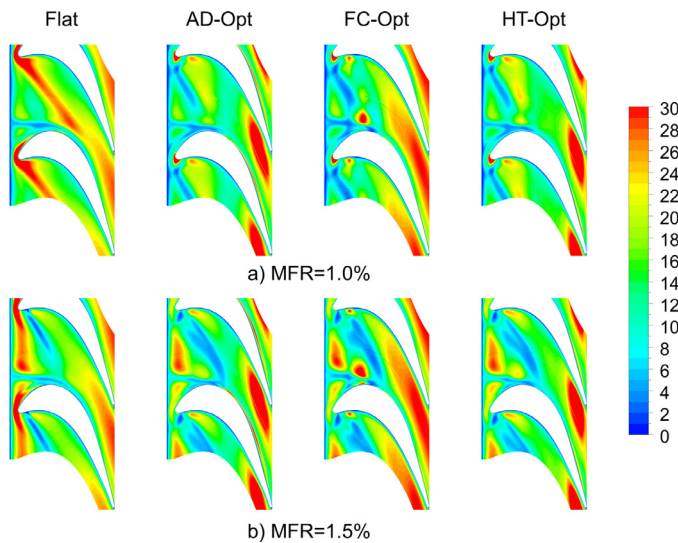
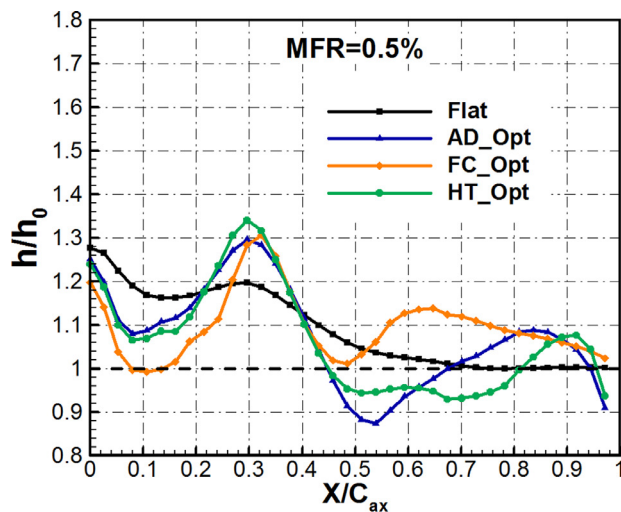


Fig. 21. Distributions of turbulent-to-laminar viscosity ratio.

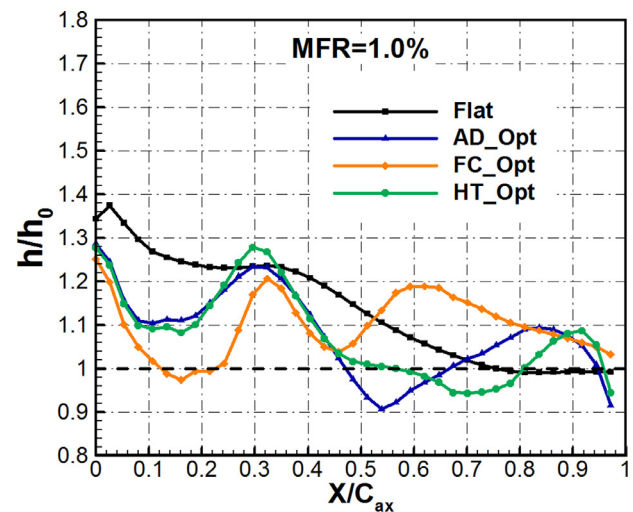
transfer there. It should be noted that the three optimal NEC designs, especially the FC-Opt, create another high heat transfer spot E, which is invisible for the flat endwall. The zone E appears just at the peak of the convex profile and is a result of flow accelerations.

It is also important to note that the heat transfer in the low-heat-transfer zone D is also strongly influenced by the NEC designs. The three optimal NEC designs, especially the FC-Opt, can effectively enlarge the area of zone D. This can be interpreted by the reduction of local turbulent mixing intensity. Evidence has shown that the intensity of local turbulent mixing largely determines the level of convective heat transfer. The distributions of turbulent-to-laminar viscosity ratio are plotted in Fig. 21 to support this finding. With the application of the NEC designs, the turbulent viscosity in the low-heat-transfer region D is substantially reduced, indicating that the turbulent mixing intensity here is considerably weakened. This is due in large part to the weakening of HVPS.

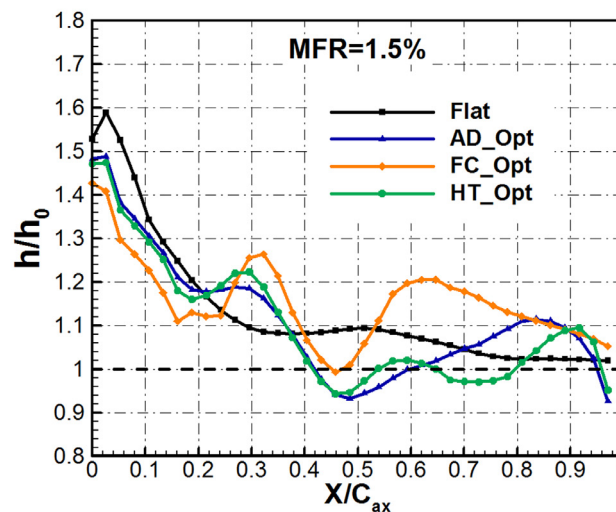
Fig. 22 quantitatively compares the axial distributions of the transversely averaged heat transfer coefficient ratio  $h/h_0$ . It should be pointed out that  $h$  and  $h_0$  mean the heat transfer rate computed with and without the purge coolant, respectively. Overall, the HT-Opt performs best in reducing the transverse average heat transfer



a) MFR=0.5%



b) MFR=1.0%



c) MFR=1.5%

Fig. 22. Axial distributions of the transversely averaged HTC.

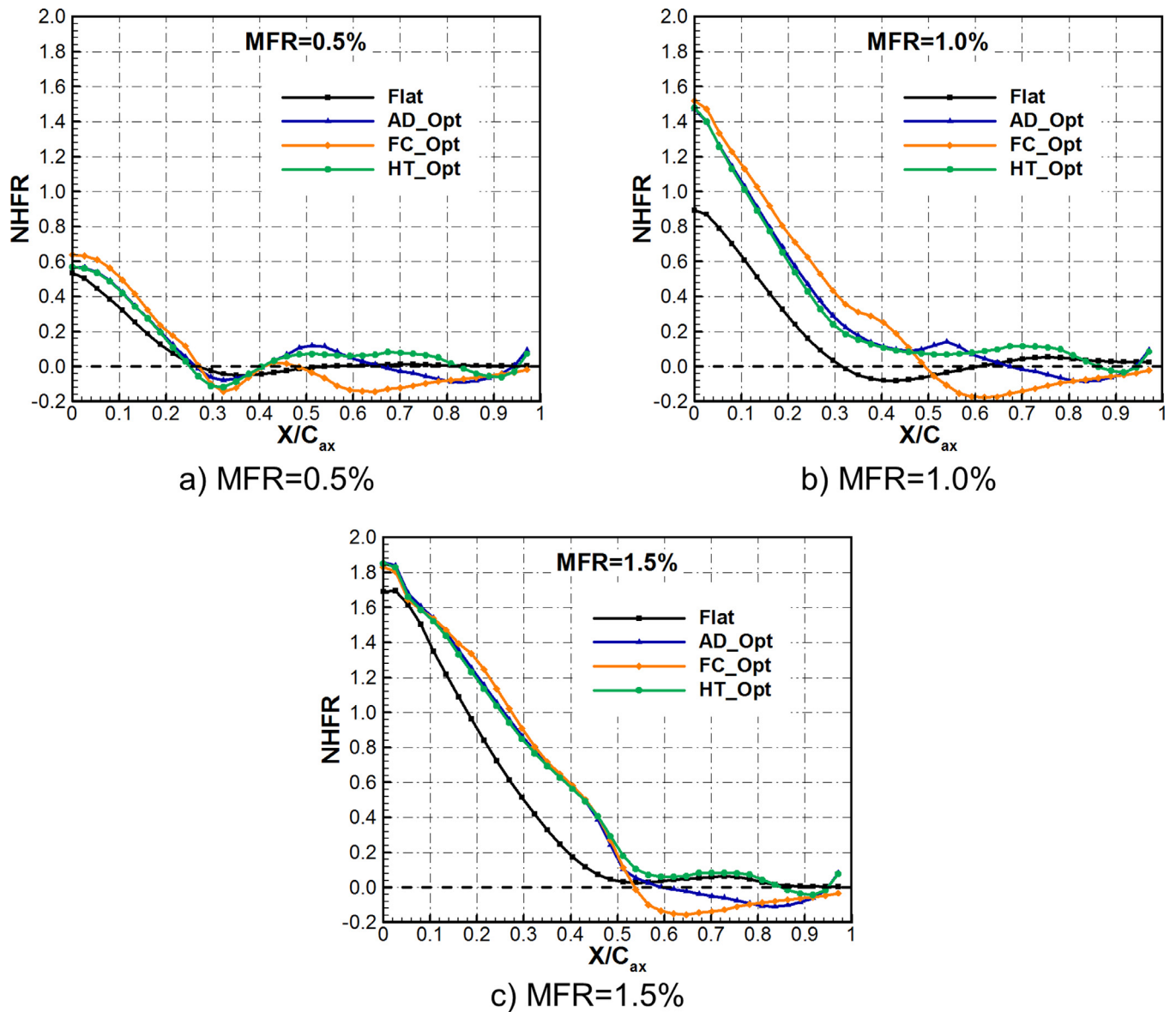


Fig. 23. Axial distributions of the transversely averaged NHFR.

intensity at all MFRs. In comparison to a flat endwall, the HT-Opt is highly effective in lowering the heat transfer in most of the areas, except for some localized areas (zones E and  $C_2$ ). The reasons for the heat transfer enhancement in the zone E and  $C_2$  have been discussed above. The changes in heat transfer due to the AD-Opt are generally similar to the HT-Opt, with only two minor differences. Firstly, the range of the high-heat-transfer zone  $C_2$  of the AD-Opt ( $0.7 < X/C_{ax} < 0.95$ ) extends further upstream than that of the HT-Opt ( $0.8 < X/C_{ax} < 0.95$ ). This is because that the AD-Opt has a longer ascending ramp, allowing the fluid to accelerate from a more upstream position. Secondly, the heat transfer in zone  $C_1$  is reduced more significantly by the AD-Opt since it has a much larger concave amplitude there. In contrast, the FC-Opt can reduce the heat transfer level of the fore-part of the endwall ( $X/C_{ax} < 0.25$ ) and the reduction is more significant than the other two designs. Unfortunately, the FC-Opt instead intensifies the heat transfer intensity of the second half of the endwall ( $X/C_{ax} > 0.55$ ).

The net heat flux reduction (NHFR for short) which integrates the film cooling and heat transfer outcomes, is commonly used to evaluate the thermal load on the endwall. The theoretical formula

of NHFR is defined in Eq. (11).

$$NHFR = 1 - \frac{h}{h_0} (1 - \eta\phi) \quad (11)$$

$$\phi = \frac{T_\infty - T_c}{T_\infty - T_w} \quad (12)$$

where  $\phi$  represents the non-dimensional metal temperature, which is defined in Eq. (12). According to Ref. [39], for a fully cooled turbine blade,  $\phi$  is usually around 1.6. It can be concluded that a higher NHFR means a lower thermal load on the endwall.

Fig. 23 quantitatively compares the axial distribution of the transversely averaged NHFR. Overall, the effect of the three optimal NEC designs on the endwall thermal load is significant, especially when  $MFR > 1.0\%$ . Their effect on the thermal load on the first half of the endwall ( $X/C_{ax} < 0.5$ ) is roughly consistent and also favorable, while the effect on the second half of the endwall ( $X/C_{ax} > 0.5$ ) is quite different. Among the three NEC designs, the FC-Opt achieves the most significant reduction in thermal load on the fore part of the endwall. However, it increases the thermal load on the rear part of the endwall compared to a flat endwall. The AD-Opt and



HT-Opt perform relatively similarly in the fore part of the endwall ( $X/C_{ax} < 0.6$ ), with the AD-Opt being slightly better. For the thermal load on the rear part of the endwall ( $X/C_{ax} > 0.6$ ), the HT-Opt performs better than the AD-Opt. Taken together, the HT-Opt achieves the most well-balanced performance for the entire endwall. It can significantly reduce the thermal load on most of the endwall, including the junction of blade LE and endwall, while only slightly increasing the local thermal load near the blade throat. In summary, the HT-Opt is expected to be a promising scheme for end-wall thermal management.

Taken together, the full-endwall non-axisymmetric contouring method proposed in this paper is demonstrated to be effective in improving the aerodynamic and thermal performance of a turbine endwall. Through an aero-thermal optimization, we obtain three representative optimal solutions, including designs that significantly improve both aerodynamic performance and endwall thermal management level. Through an in-depth analysis of flow and thermal details, we elucidate the mechanism of different NEC designs on the aero-thermal performance and summarize design guidelines that are favorable to improve endwall performance. More specifically, the endwall convex profile near the blade LE is a common feature that contributes beneficially to aerodynamics, film cooling and heat transfer. The design of the endwall profile between the adjacent blades is more complicated due to the clear trade-offs between the different design objectives. Therefore, the gas turbine designers are suggested to design the NEC very delicately, with a reasonable balance of different objectives. Aero-thermal optimization is a promising way to obtain desirable NEC schemes.

## 5. Conclusions

This paper proposes a novel and practicable non-axisymmetric endwall contouring approach. The effectiveness of the proposed NEC is explored on a stationary gas turbine blade with a typical purge slot. To better find the desirable solutions, three separate optimizations of the NEC are performed with the objectives of minimum aerodynamic loss, highest film cooling effectiveness and lowest heat transfer coefficient. The influences of the typical optimal designs on endwall aero-thermal performance are analyzed, and the underlying flow-thermal fundamentals are also elucidated in depth. The main conclusions are drawn as follows:

- (1) Through the aero-thermal optimizations, we obtain three representative optimal solutions. The FC-Opt shows substantial improvements in film cooling and heat transfer performance, at the cost of slightly higher aerodynamic losses. Specifically, the FC-Opt significantly enhances the area-averaged FCE by 8.3%, 65.2%, and 25.1% at MFRs of 0.5%, 1.0%, and 1.5%, respectively. In contrast, the AD-Opt and the HT-Opt can effectively improve all aspects of the endwall aero-thermal performance due to the weakening of the horseshoe vortex and the cross flows. Among them, the HT-Opt is the most desirable solution since it not only effectively lowers aerodynamic losses but also dramatically reduces thermal loads of almost the entire endwall.
- (2) The non-axisymmetric endwall profiles near the blade LE are quite similar for the three optimal designs. It is indicated that a convex profile near the blade LE is beneficial for all the three performance indicators. This is because that this convex profile can effectively weaken the horseshoe vortex and its two branches and diminish the turbulent mixing near the blade LE. As a result, this convex profile lowers the aerodynamic losses and improves the cooling effectiveness while reducing the heat transfer intensity.
- (3) The three optimal designs differ in the endwall profile between the two adjacent blades. In general, the AD-Opt and the HT-Opt

are somewhat similar in endwall profiles. The AD-Opt and the HT-Opt have distinct valleys near the rear part of the blade suction side. Such concave profiles help to reduce the transverse pressure gradient, thus weakening the cross flows and the passage vortex. As a result, the aerodynamic losses and heat transfer coefficients of the two optimal designs are effectively reduced. Unlike the above two designs, the FC-Opt has a significant convex profile near the middle of the blade suction side. This distinct convex profile accelerates the coolant and allows it to adhere to the endwall surface, thus enhancing the film cooling coverage at the front endwall. In summary, the design of non-axisymmetric endwall between the two adjacent blades is complicated due to the trade-offs between different objectives.

Overall, this paper proposes an efficient non-axisymmetric end-wall contouring approach and also elucidates its fundamental physics, which is expected to provide helpful guidelines for gas turbine designers.

## Declaration of Competing Interest

None

## CRediT authorship contribution statement

**Zhi Tao:** Conceptualization, Methodology, Investigation, Writing – original draft. **Zhendong Guo:** Investigation, Writing – review & editing. **Boyang Yu:** Visualization, Data curation. **Liming Song:** resources, Software. **Jun Li:** Supervision.

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